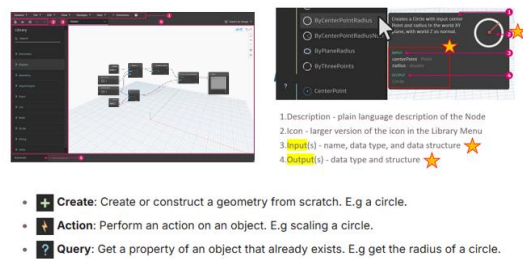


Day	Time	Topic	Detailed Description
1	1	Dynamo & Generative Design Trends	활용 사례 소개 및 목표
	2	Dynamo 개념 및 인터페이스	Dynamo, Generative Design 개념, 작동 방식 및 UI 확인
	3	Dynamo 기본 Geometry 활용	Point, Line, PolyCurve, Surface, Solid 순환
	4	Geometry 응용	응용 및 Revit 정보 및 형상 연결
2	1	공간 배치 자동화 개념 소개	다양한 Space Planning, Space Layout 사례 및 방식 설명
	2	공간 배치 자동화 예제 1	공간 배치를 활용한 형상 생성, 고정된 형상의 공간 배치
	3	공간 배치 자동화 예제 2	Space Planning - 주거 공간 배치 예제
	4	공간 배치 자동화 예제 3	형상 생성을 통한 1~4개 건물 규모 검토 (건폐율, 용적률)
3	1	주차장 배치 및 검토 자동화 개념 소개	주차장 배치 자동화 개념 및 사례 소개
	2	주차장 배치 검토 조건 확인	Revit 패밀리 활용 / Dynamo Geometry 활용 비교
	3	외부 주차 배치 방식 개념	정해진 공간의 외부 주차 배치 방식 개념 확인
	4	내부 주차 배치 방식 개념	정해진 공간의 외부를 제외한 주차 배치 개념 확인
4	1	공원 구성요소 배치 자동화 개념 소개	공원 진입로의 변경과 휴식 공간 검토 사례
	2	공원 구성요소 배치 자동화 실습	정해진 공간의 구성요소를 변경하며 배치 검토 실습
	3	외피 패턴 형성 자동화 개념 소개	Facade 패턴 변경 사례
	4	외피 패턴 형성 자동화 실습	다양한 노드를 활용한 패턴 변경 방식 실습
	5	Dynamo 별도 실행	Sandbox 설치 및 Curve Mapper Node 확인
			Curve Mapper Example
			Geometry SAT 내보내기
			SAT Revit 형상 적용

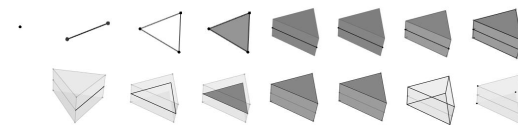
Day	Time	Topic	Detailed Description
1	1	Dynamo & Generative Desgin Trends	활용 사례 소개 및 목표
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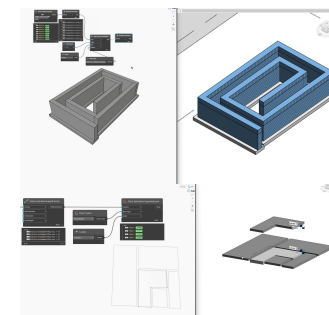
**Dynamo &
Generative Design Trends**



**Dynamo Concept
Interface**



Geometry Basic



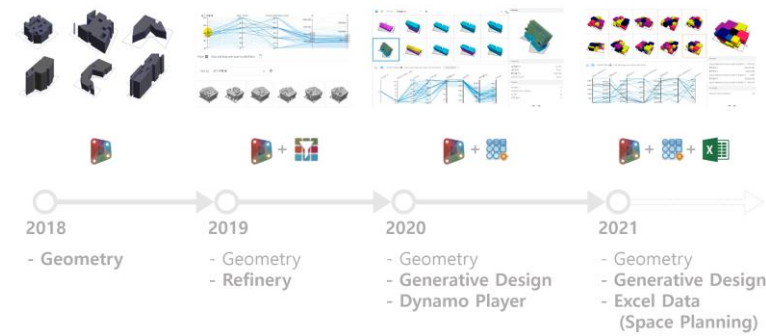
Geometry - Revit

01 Dynamo & Generative Design Trends

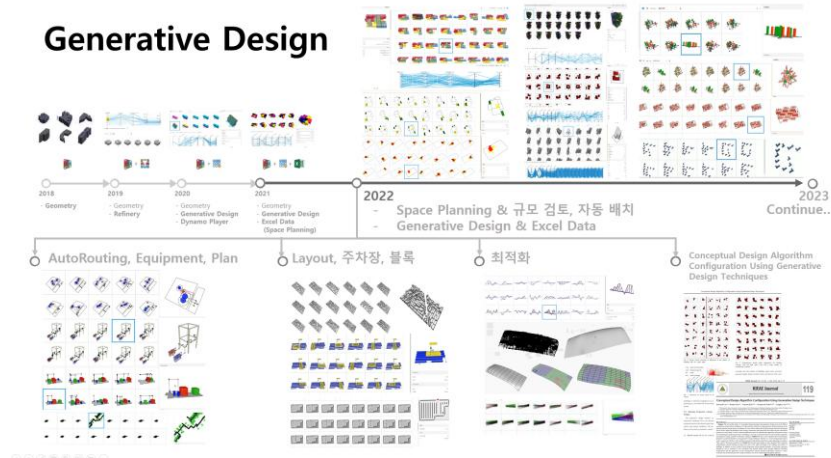
활용 사례 소개 및 목표

Generative Design

Space Planning

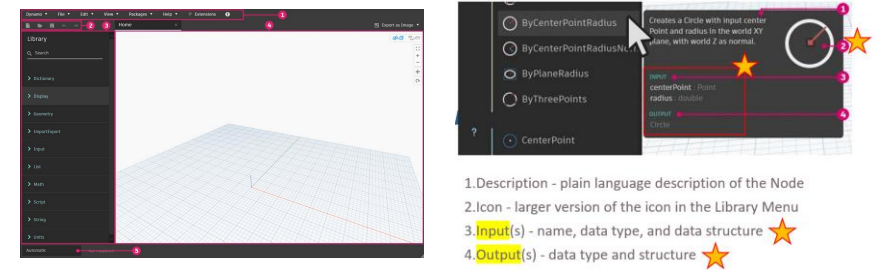


Generative Design

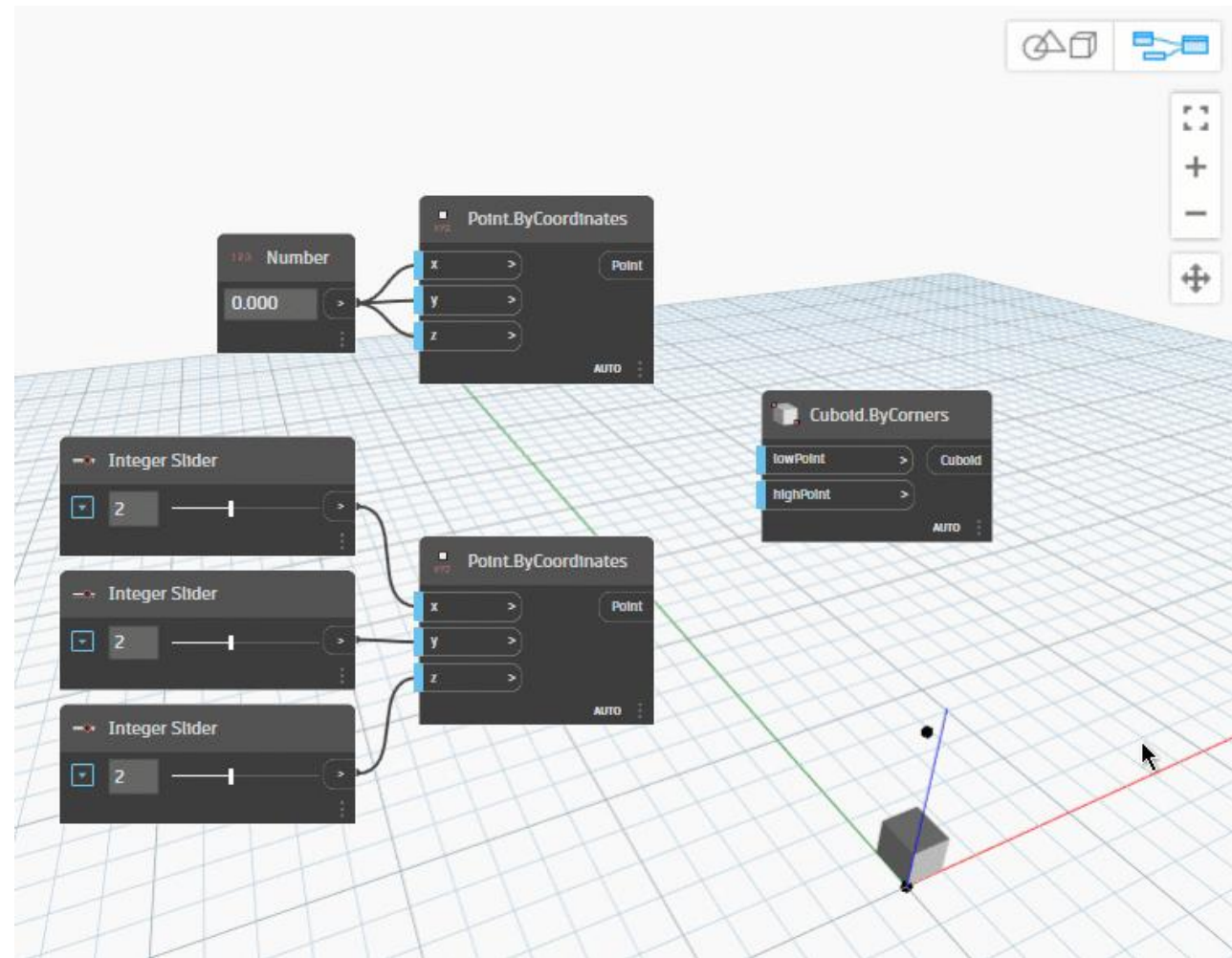
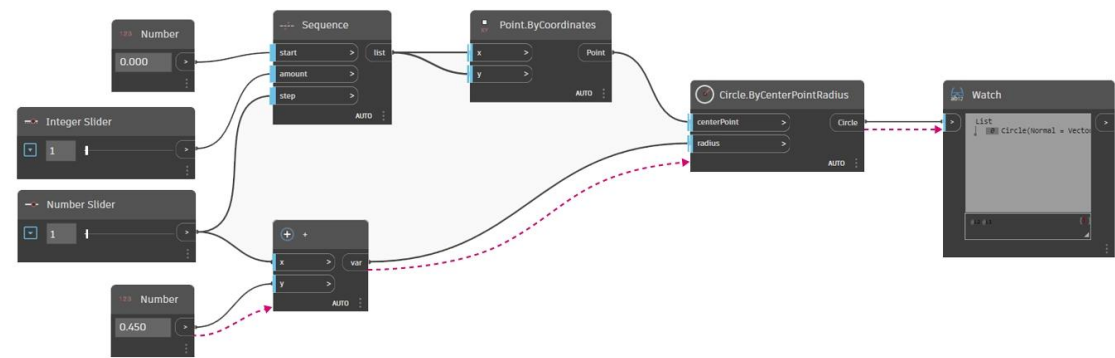


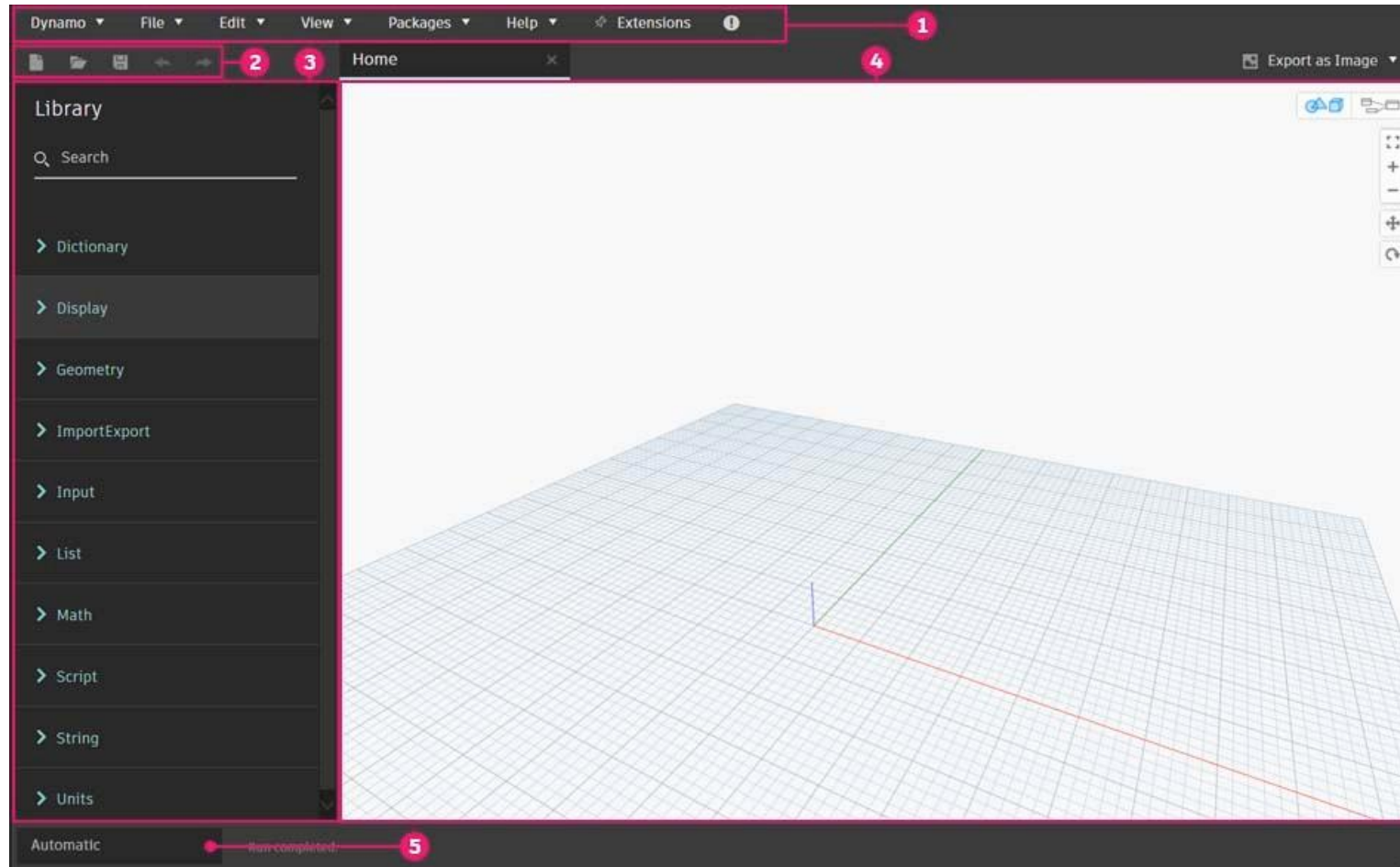
02 Dynamo Concepts and Interfaces

Dynamo, Generative Design 개념, 작동 방식 및 UI 확인

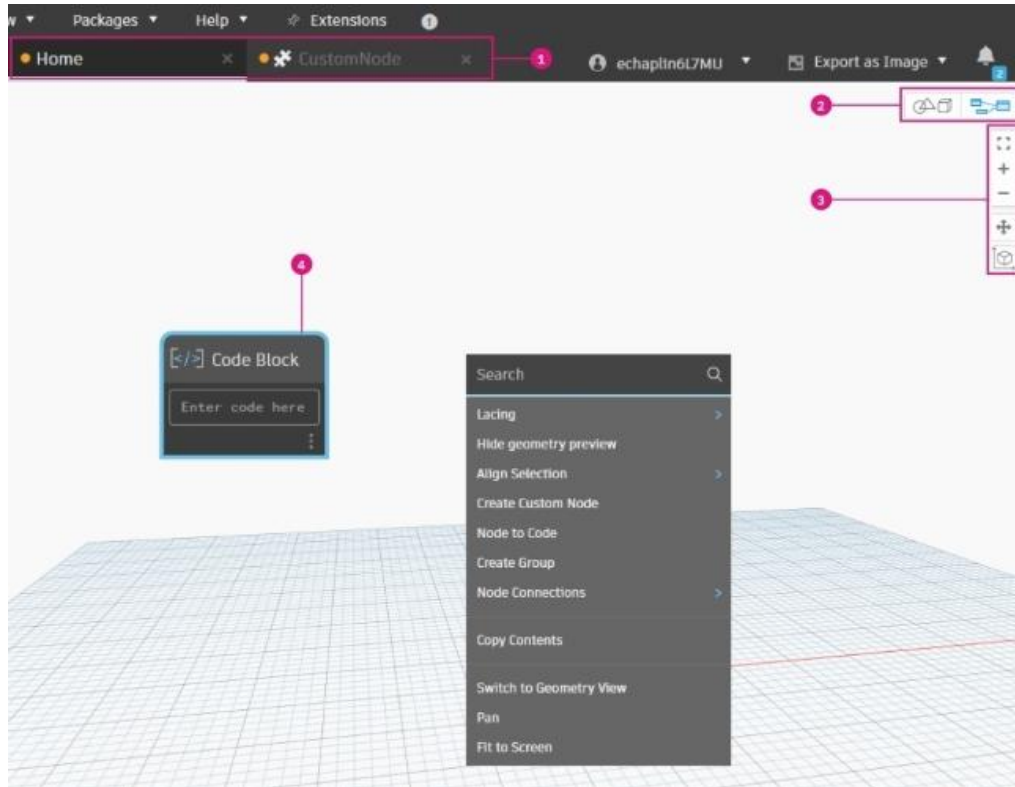


- + Create:** Create or construct a geometry from scratch. E.g a circle.
- ⚡ Action:** Perform an action on an object. E.g scaling a circle.
- ? Query:** Get a property of an object that already exists. E.g get the radius of a circle.

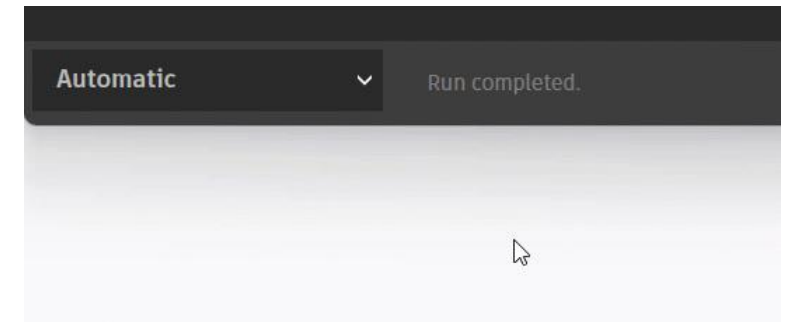
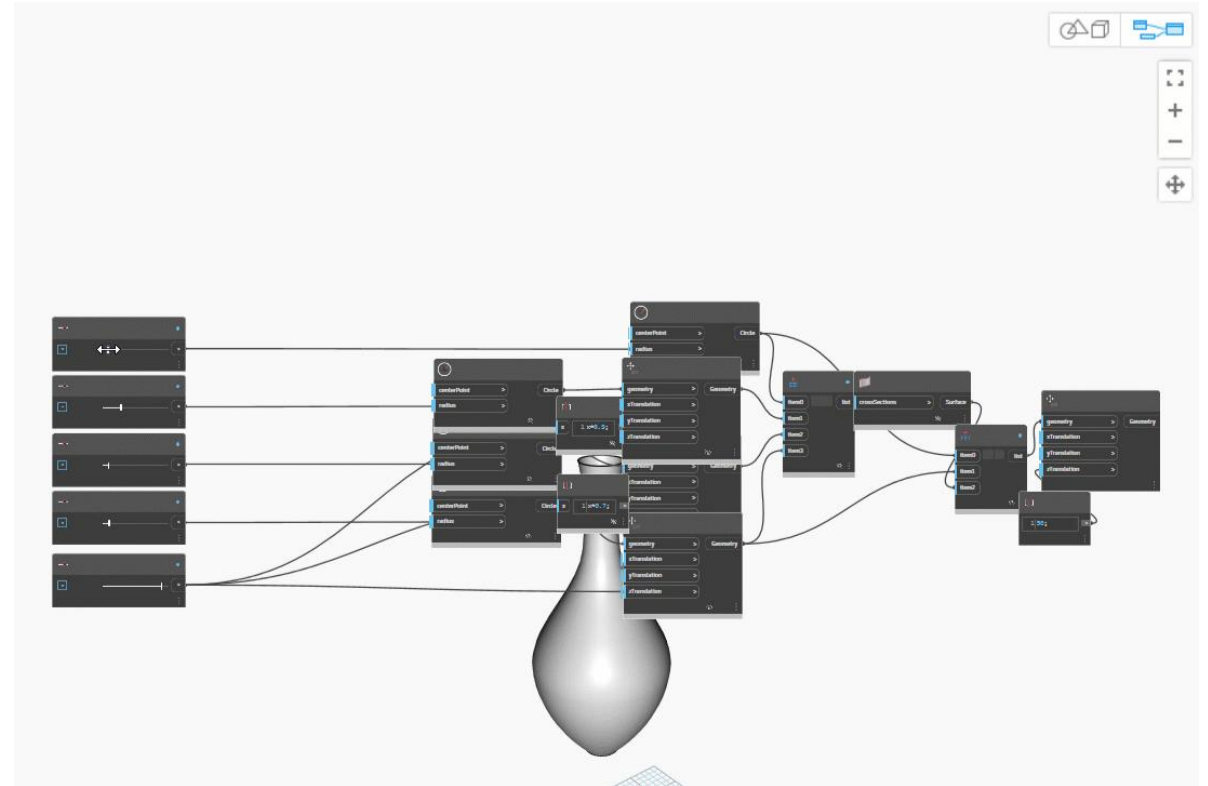


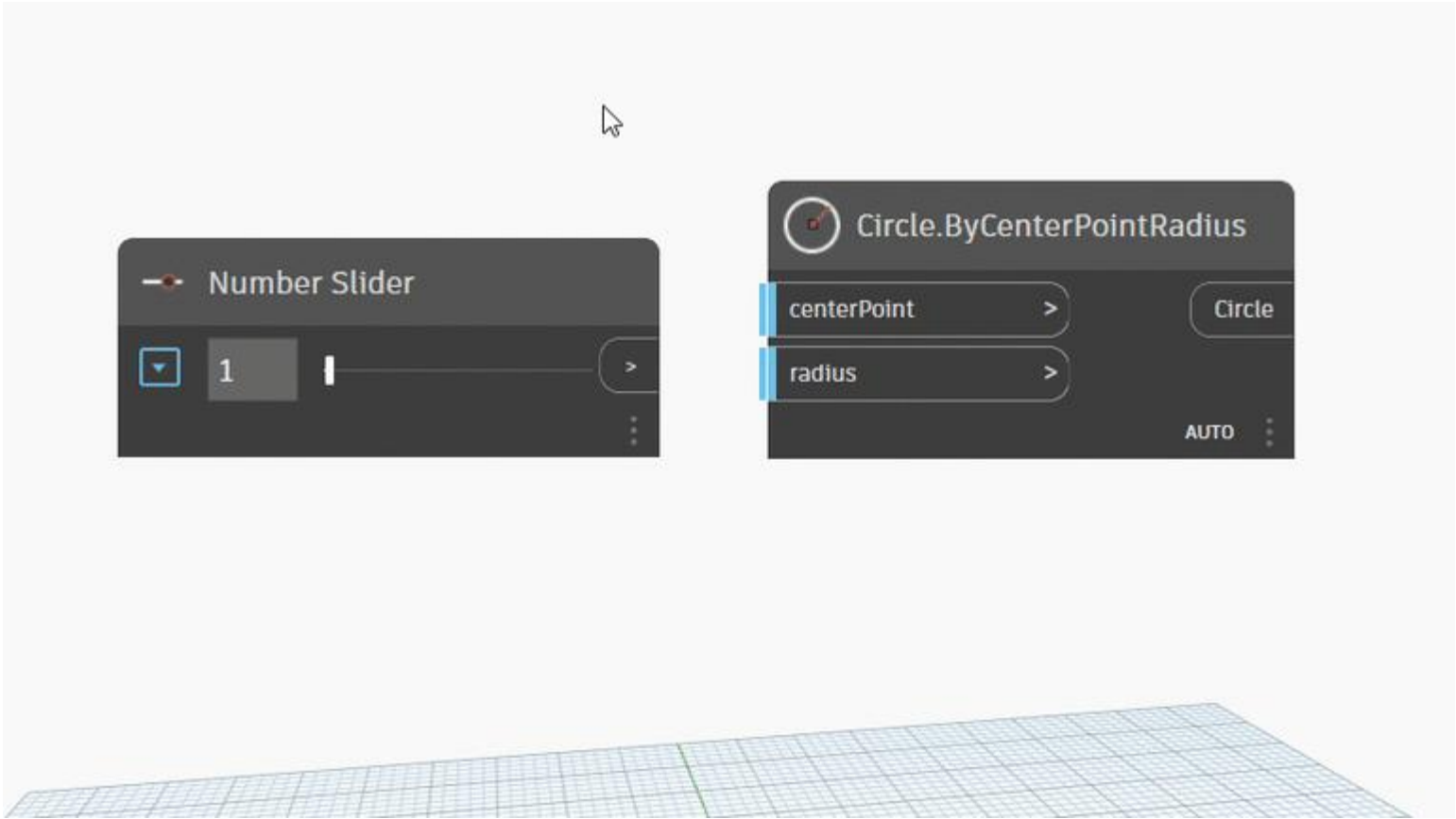


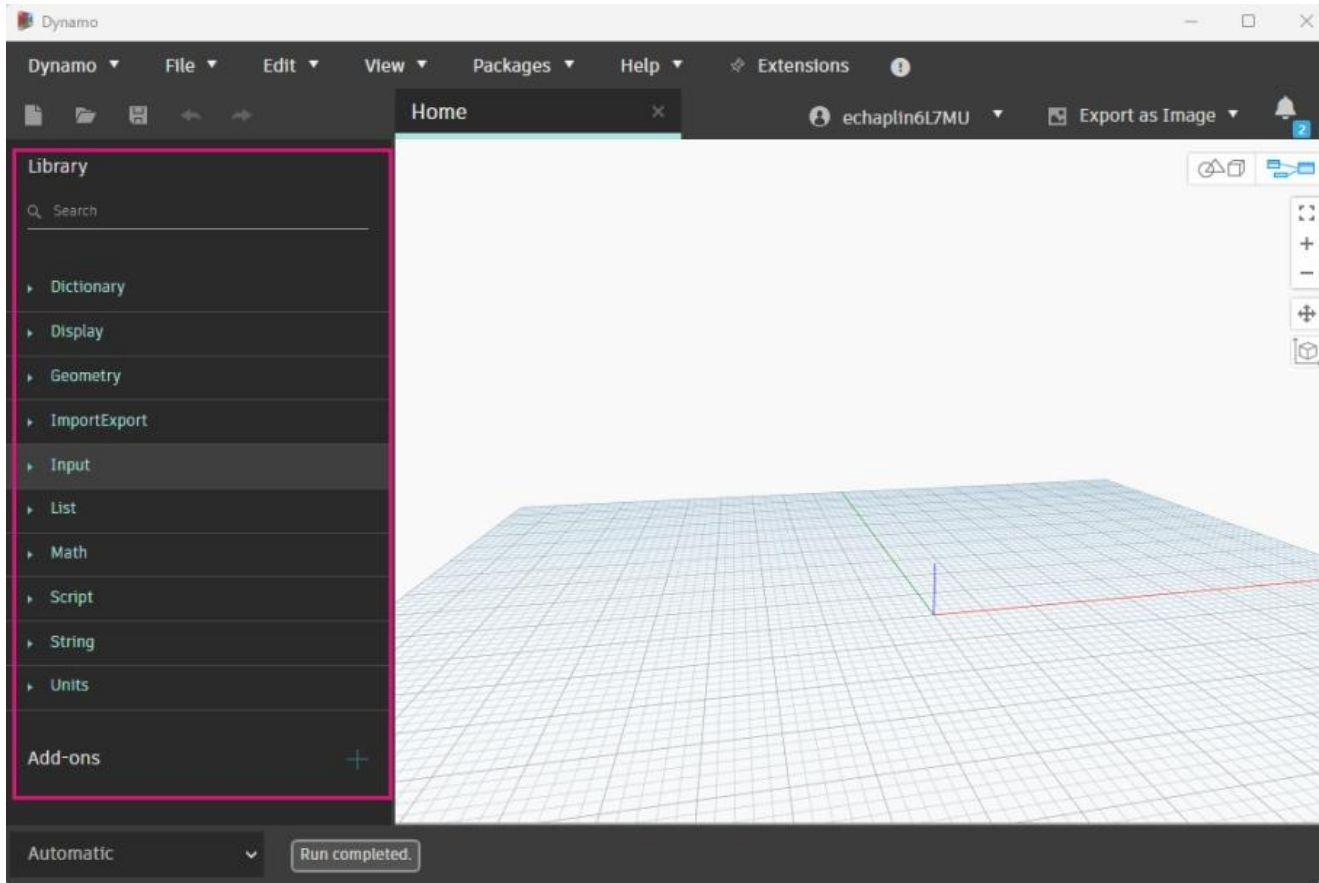
- 1.Menu
- 2.Toolbar
- 3.Library
- 4.Workspace
- 5.Execution bar



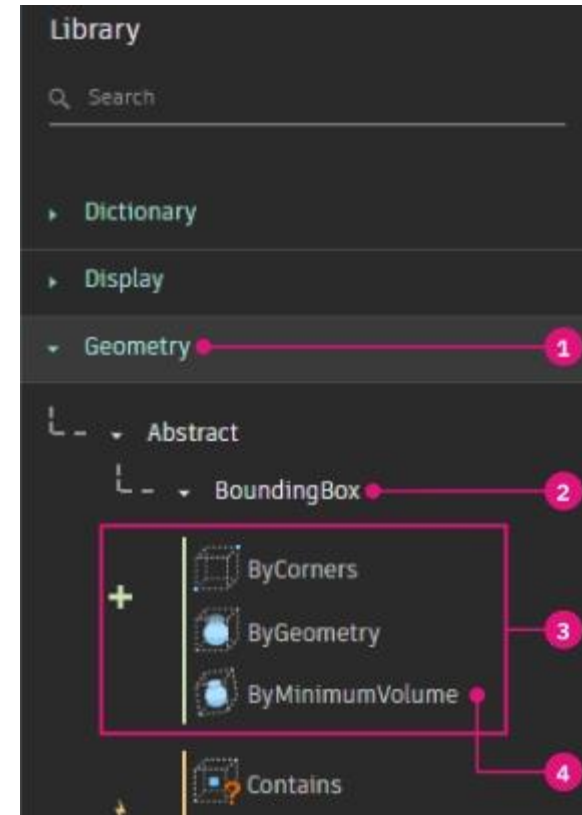
- 1.All Active Tabs.
- 2.Preview Mode
- 3.Zoom/Pan Controls
- 4.Node in Workspace



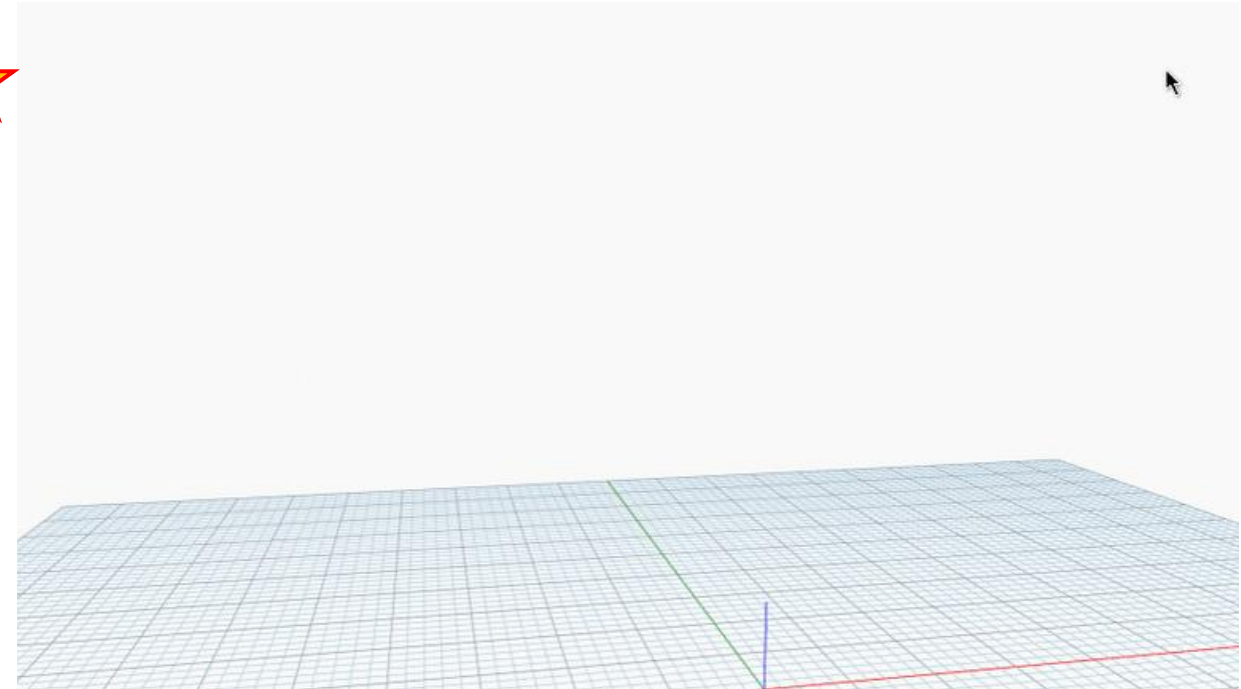
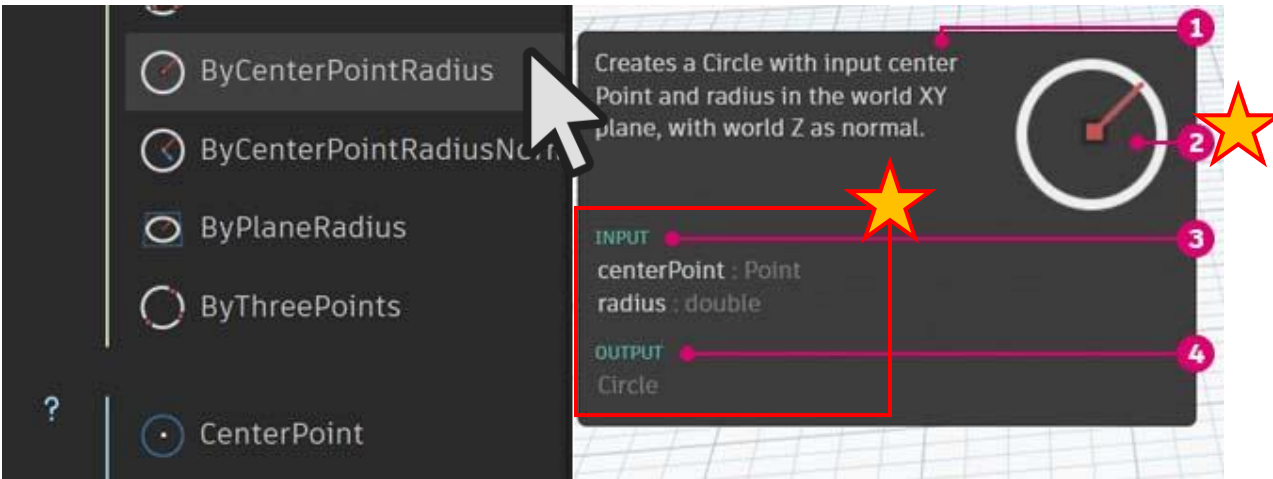




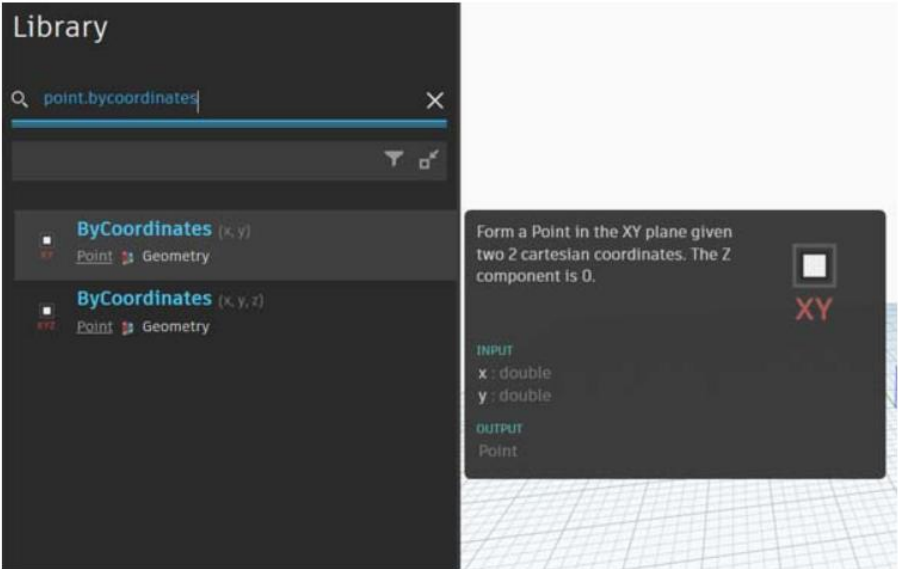
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- **? Query:** Get a property of an object that already exists. E.g get the radius of a circle.



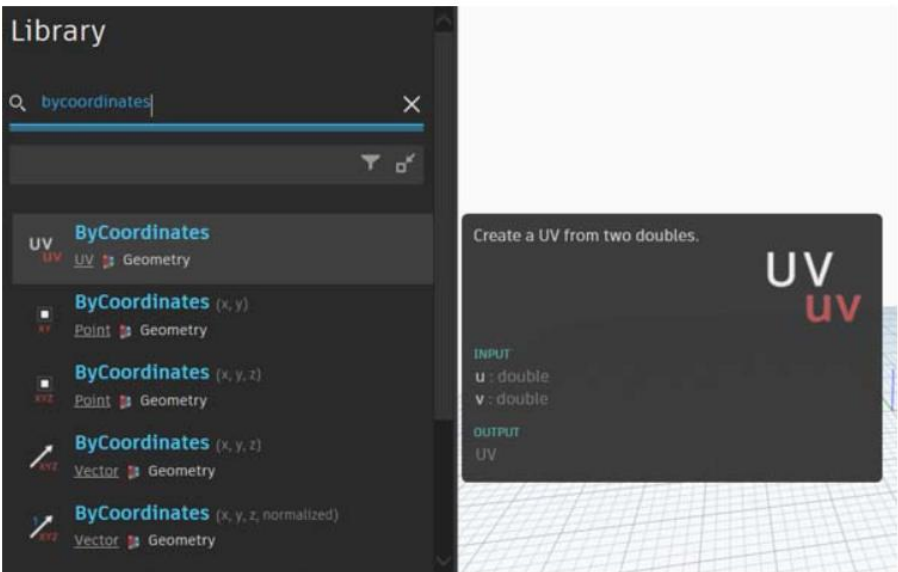
1. Library
2. Category
3. Subcategory
4. Node



- `category.nodeName`

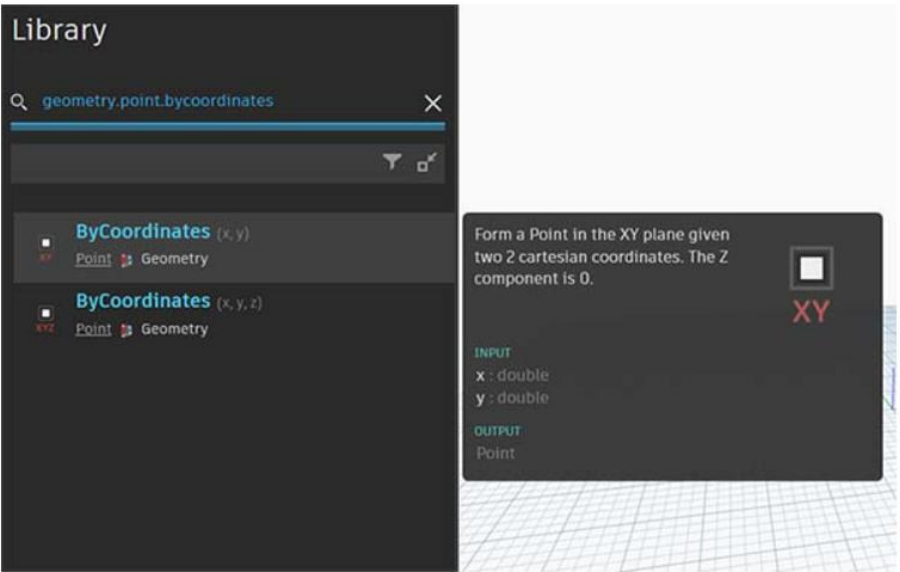


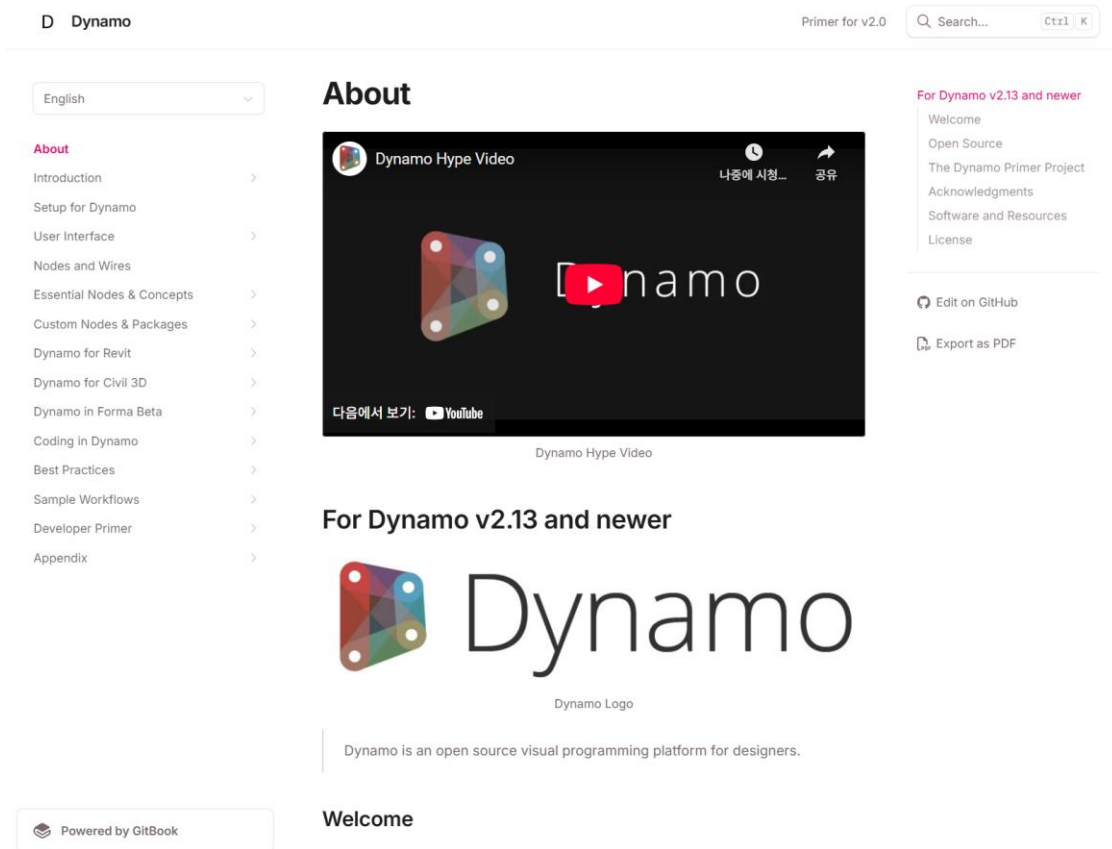
- `nodeName` or `keyword`



Typing in different portions of the Node's place in the Library hierarchy in the `library.category.nodeName` format returns different results

- `library.category.nodeName`



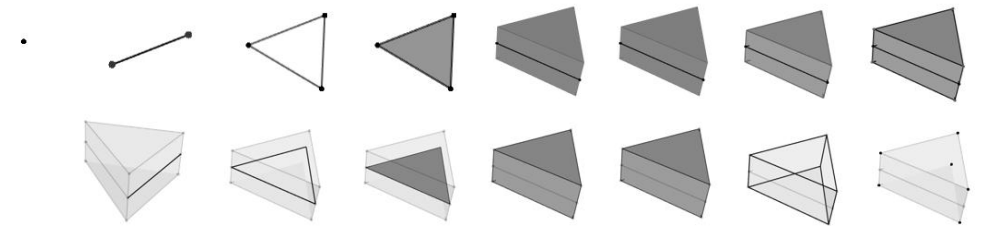


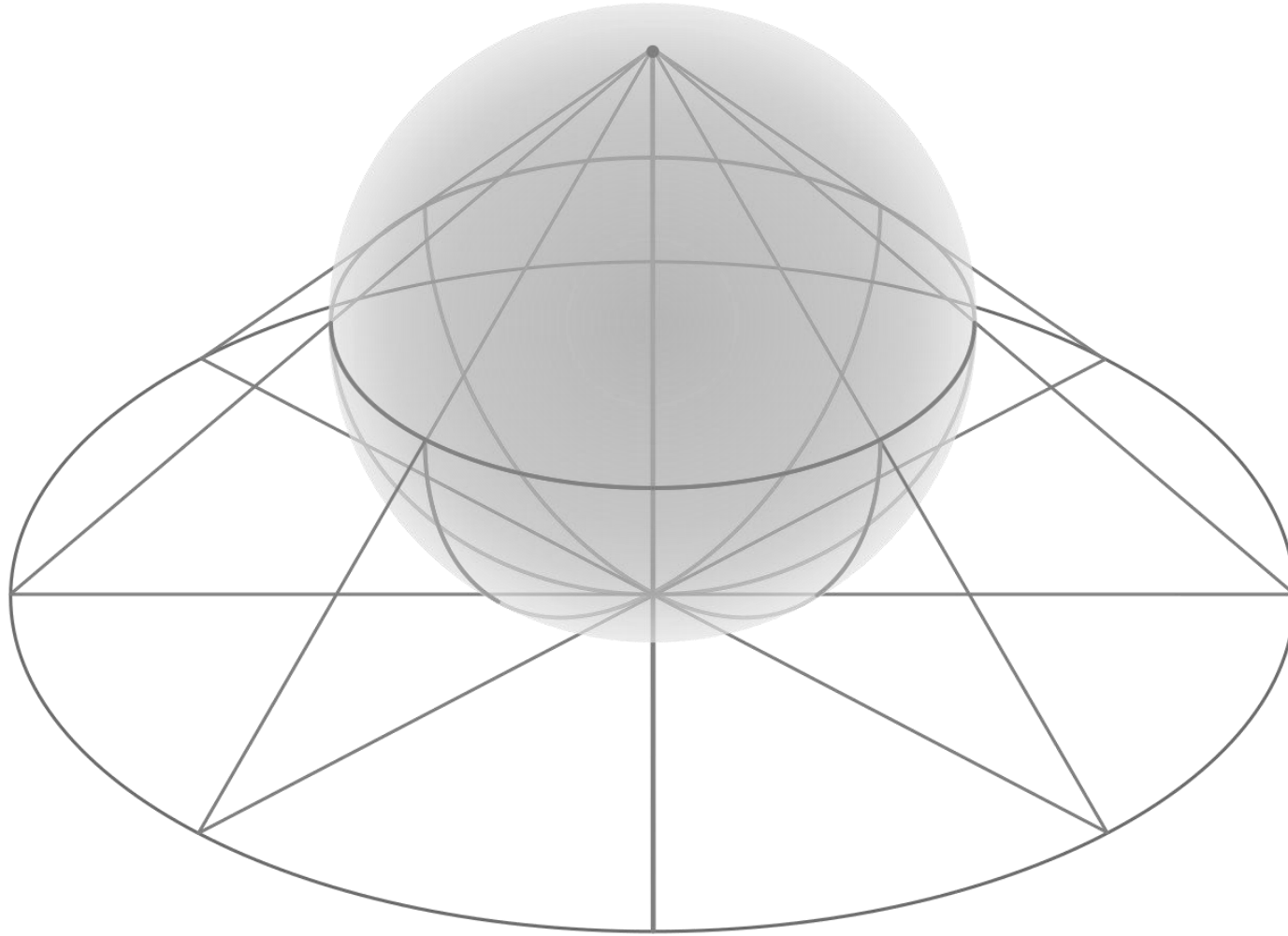
https://primer2.dynamobim.org/4_nodes_and_wires

https://primer.dynamobim.org/

03 *Dynamo Basic [Geometry]*

Point, Line, PolyCurve, Surface, Solid 순환



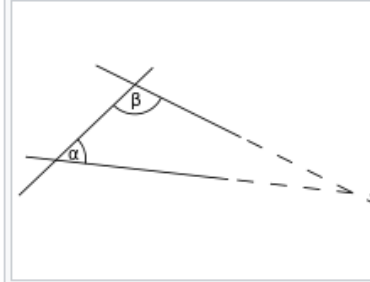




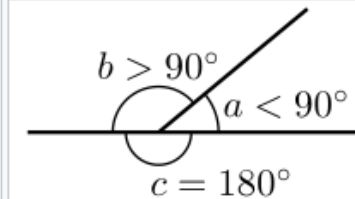
A European and an Arab practicing geometry in the 15th century



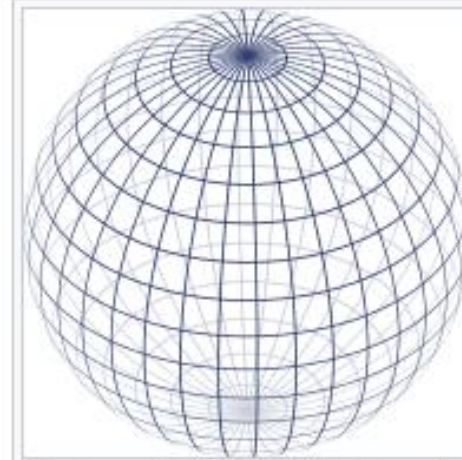
Woman teaching geometry. Illustration at the beginning of a medieval translation of Euclid's *Elements*, (c. 1310).



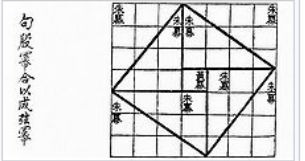
An illustration of Euclid's parallel postulate



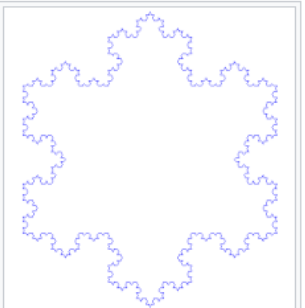
Acute (a), obtuse (b), and straight (c) angles. The acute and obtuse angles are also known as oblique angles.



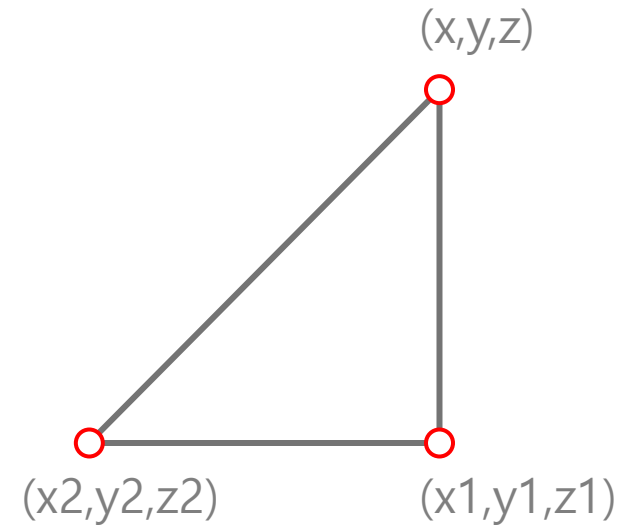
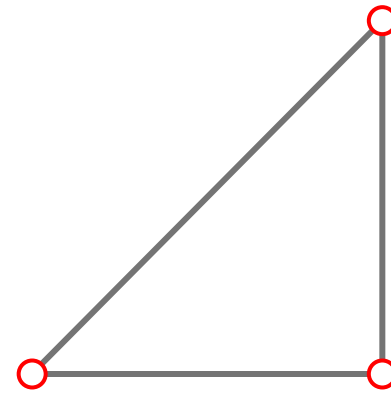
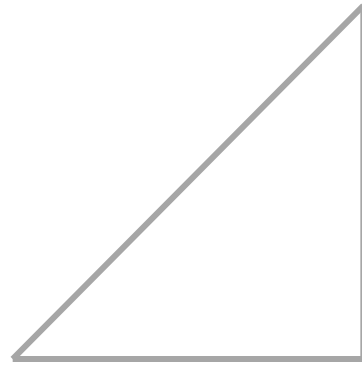
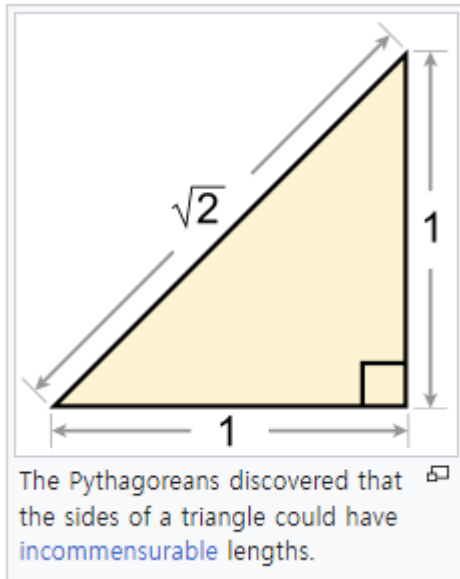
A sphere is a surface that can be defined parametrically (by $x = r \sin \theta \cos \varphi$, $y = r \sin \theta \sin \varphi$, $z = r \cos \theta$) or implicitly (by $x^2 + y^2 + z^2 - r^2 = 0$.)



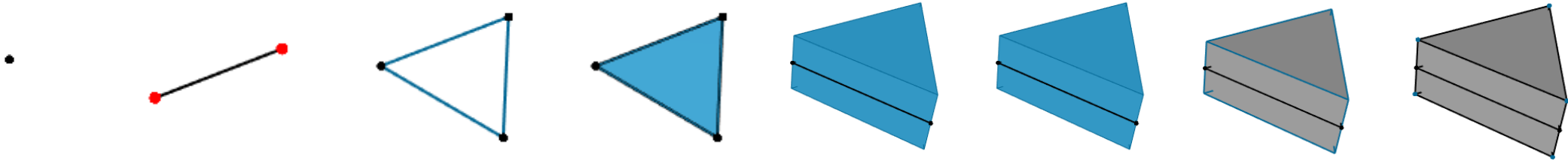
Visual checking of the Pythagorean theorem for the (3, 4, 5) triangle as in the *Zhoubi Suanjing* 500–200 BC. The Pythagorean theorem is a consequence of the Euclidean metric.



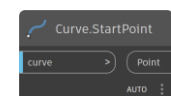
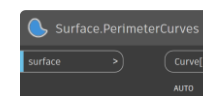
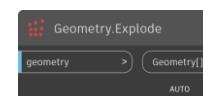
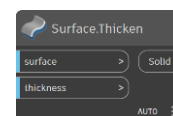
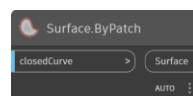
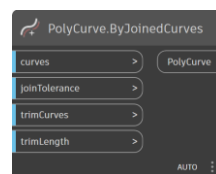
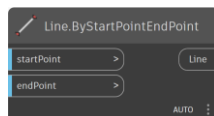
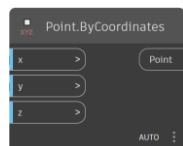
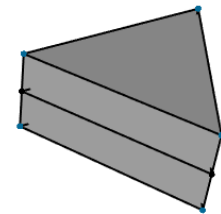
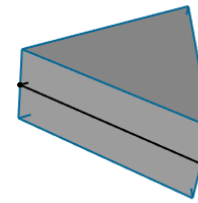
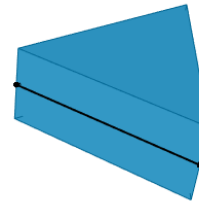
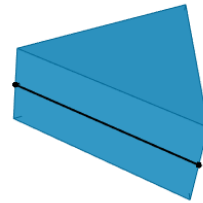
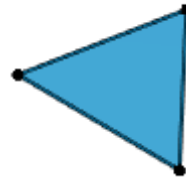
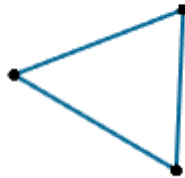
The Koch snowflake, with fractal dimension $= \log 4 / \log 3$ and topological dimension $= 1$



Point > Curve > Surface > Solid > Surface > Curve > Point

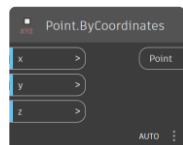


Point > Curve > Surface > Solid > Surface > Curve > Point

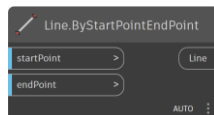


Point > Curve > Surface > Solid > Surface > Curve > Point

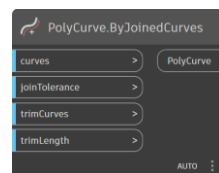
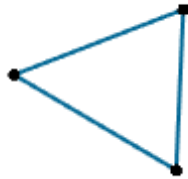
Point



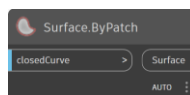
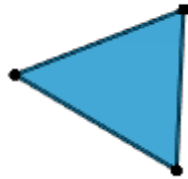
Curve
Line



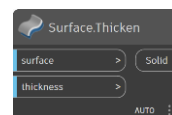
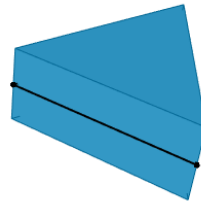
PolyCurve



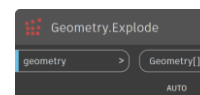
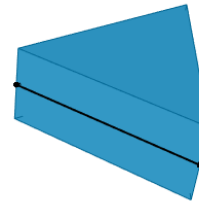
Surface



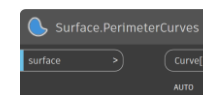
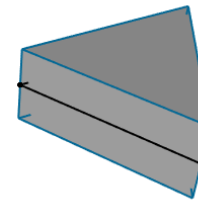
Solid



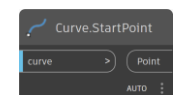
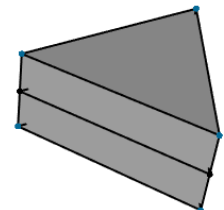
Surfaces



Curves
Lines

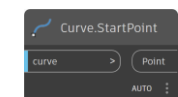
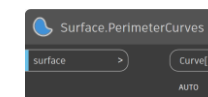
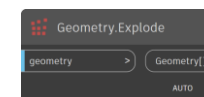
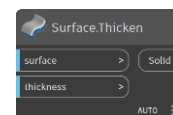
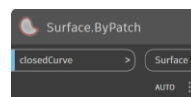
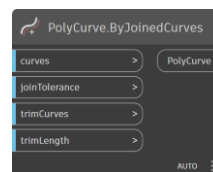
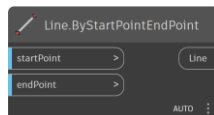
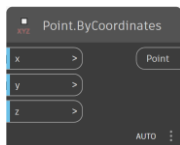
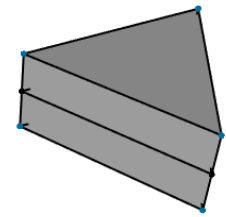
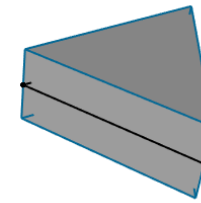
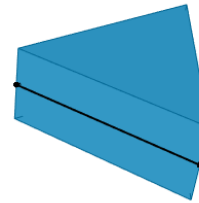
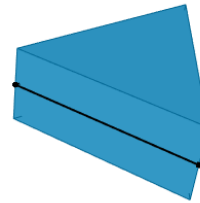
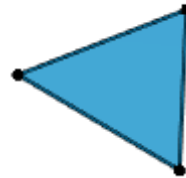
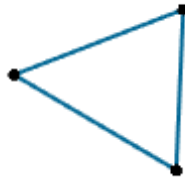


Points

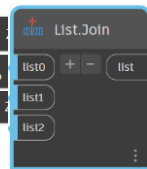


Point > Curve > Surface > Solid > Surface > Curve > Point

Point Curve
Line PolyCurve Surface Solid Surfaces Curves
Lines Points



```
Point(X = 99.412, Y = 116.216,
Point(X = 293.832, Y = 176.506,
Point(X = 258.493, Y = -6.920,
```



```
List
0 Line(StartPoint = Point(X = 99.4
1 Line(StartPoint = Point(X = 293.
2 Line(StartPoint = Point(X = 258.
@2 @1 {3}
```

PolyCurve(NumberOfCurves = 3)

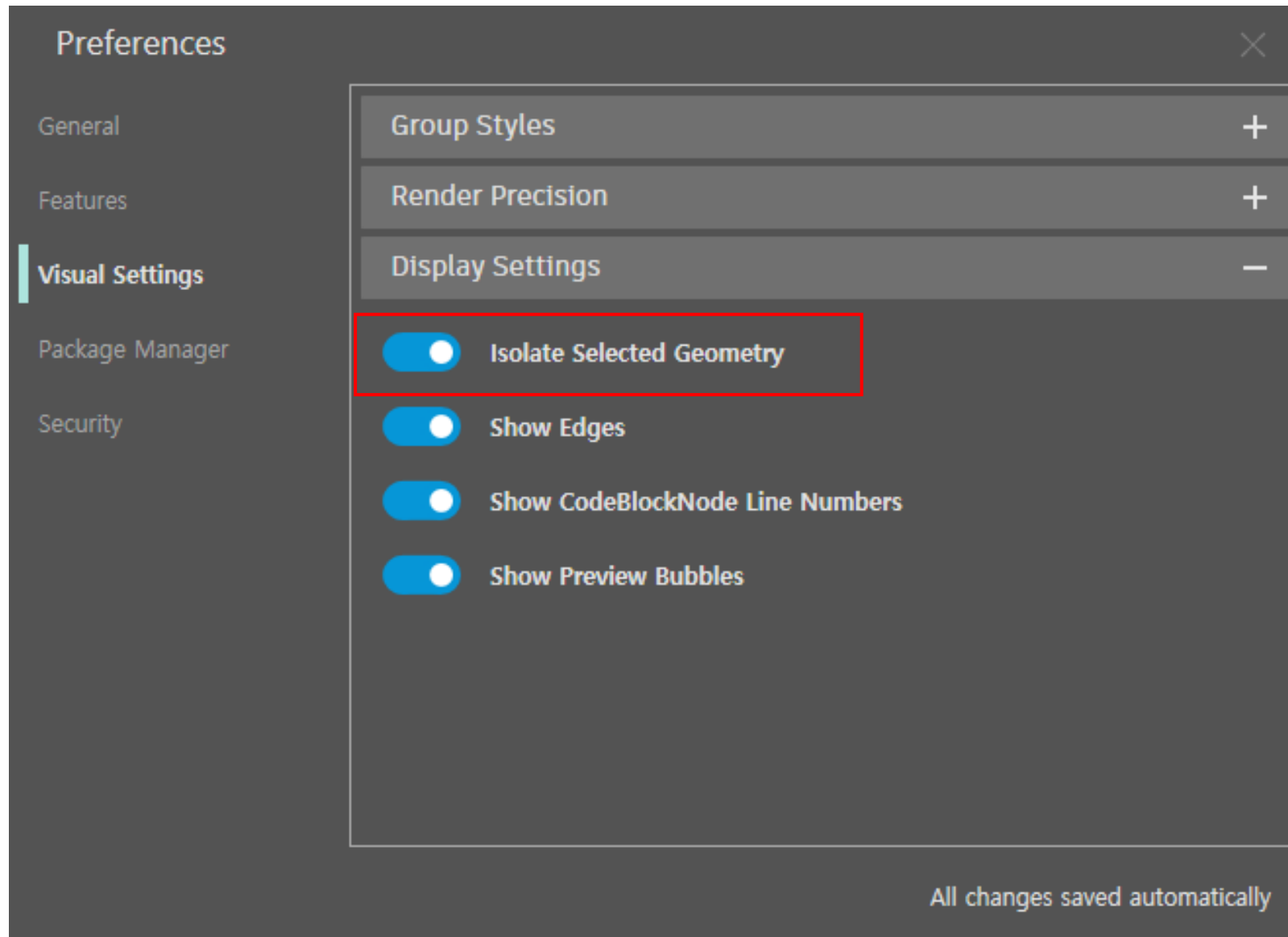
Surface

Solid

```
List
0 Surface
1 Surface
2 Surface
3 Surface
@2 @1
```

```
List
> 0 List
0 Line(StartPoint = Point(X
1 Line(StartPoint = Point(X
2 Line(StartPoint = Point(X
1 List
0 Line(StartPoint = Point(X
1 Line(StartPoint = Point(X
2 Line(StartPoint = Point(X
2 List
0 Line(StartPoint = Point(X
1 Line(StartPoint = Point(X
2 Line(StartPoint = Point(X
3 Line(StartPoint = Point(X
3 List
0 Line(StartPoint = Point(X
@3 @2 @1 {18}
```

```
List
> 0 List
0 Point(X = 99.412, Y = 116.216
1 Point(X = 293.832, Y = 176.50
2 Point(X = 258.493, Y = -6.920
1 List
0 Point(X = 99.412, Y = 116.216
1 Point(X = 293.832, Y = 176.50
2 Point(X = 293.832, Y = 176.50
2 Point(X = 99.412, Y = 116.216
3 List
0 Point(X = 293.832, Y = 176.50
@3 @2 @1 {18}
```



Point > Curve > Surface > Solid > Surface > Curve > Point

Point

Curve
Line

PolyCurve

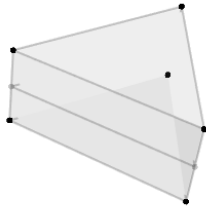
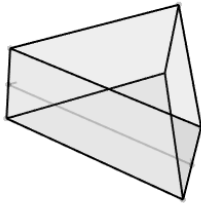
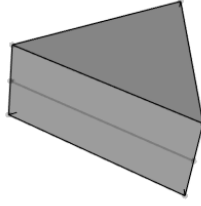
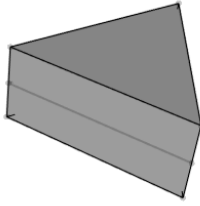
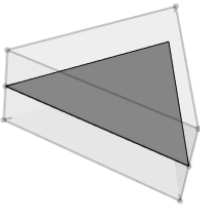
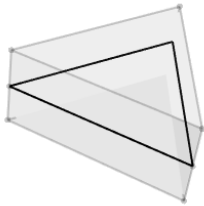
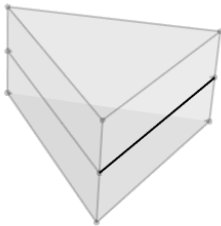
Surface

Solid

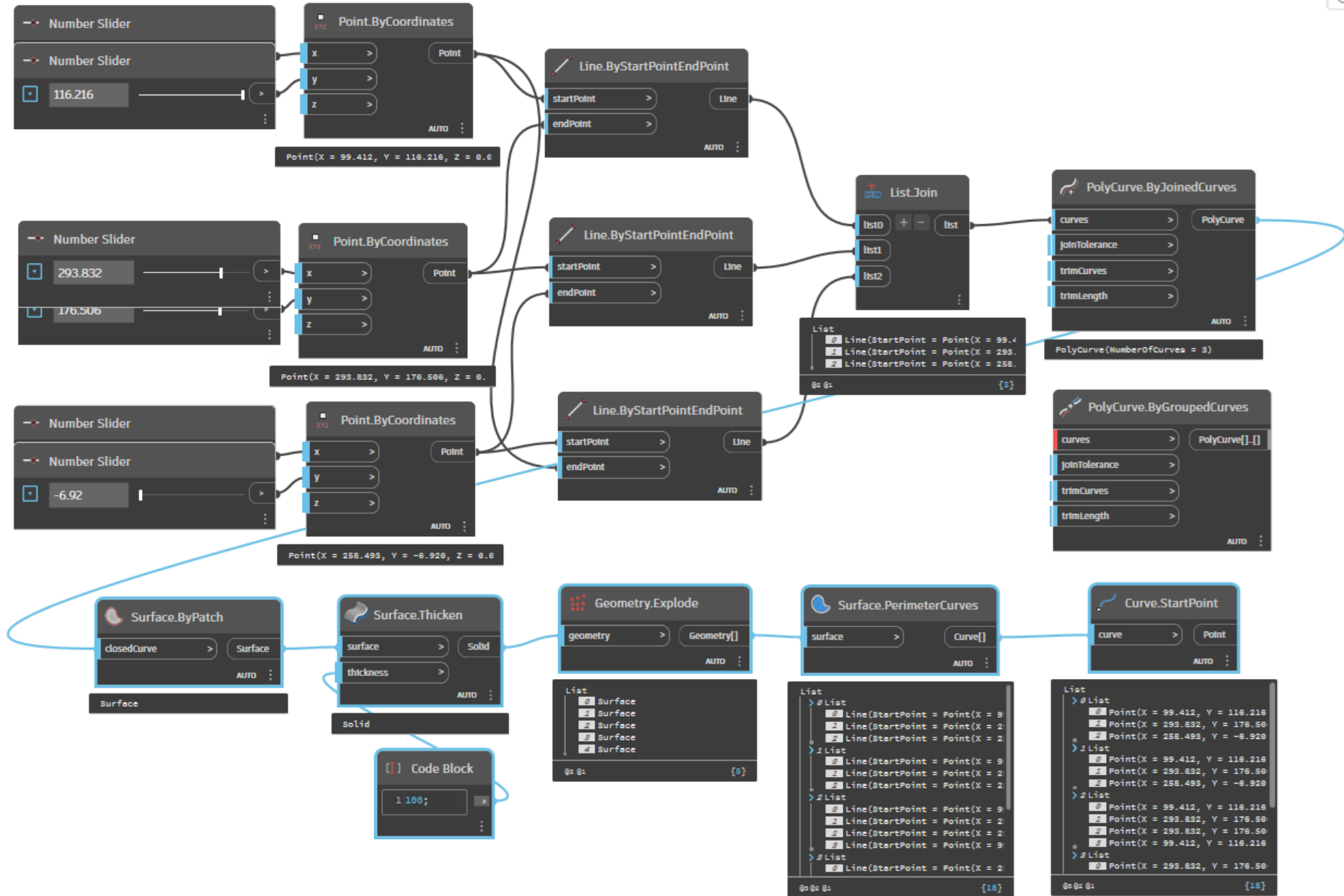
Surface

Curve
Line

Point







Geometry

Library

Search

Dictionary

Display

Geometry

Abstract

Curves

Meshes

Modifiers

Points

Solids

Surfaces

Tessellation

TSpline

Points

Point

ByCartesianCoordinates

ByCoordinates (x, y)

ByCoordinates (x, y, z)

ByCylindricalCoordinates

BySphericalCoordinates

Origin

Add

AsVector

Project

PruneDuplicates

Subtract

X

Y

Z

Curves

Arc

Circle

Curve

Ellipse

EllipseArc

Helix

Line

NurbsCurve

PolyCurve

Polygon

Rectangle

Curve

ByBlendBetweenCurves

ByIsoCurveOnSurface

ByParameterLineOnSurface

ApproximateWithArcAndLineSegment

CoordinateSystemAtParameter

CoordinateSystemAtSegmentLength

EndParameter

Extend

ExtendEnd

ExtendStart

Line

ByBestFitThroughPoints

ByStartPointDirectionLength

ByStartPointEndPoint

ByTangency

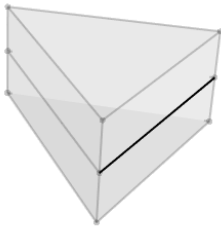
Direction

Point > Curve > Surface > Solid > Surface > Curve > Point

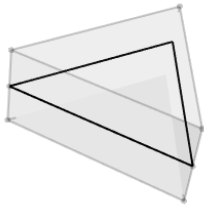
Point

?
Count

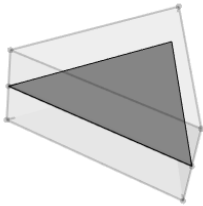
Curve
Line



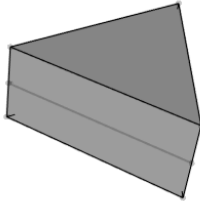
PolyCurve



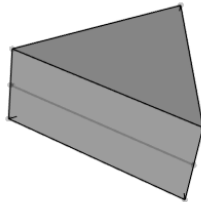
Surface



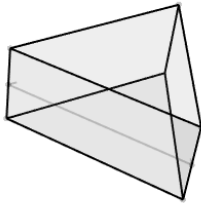
Solid



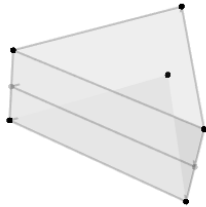
Surface



Curve
Line

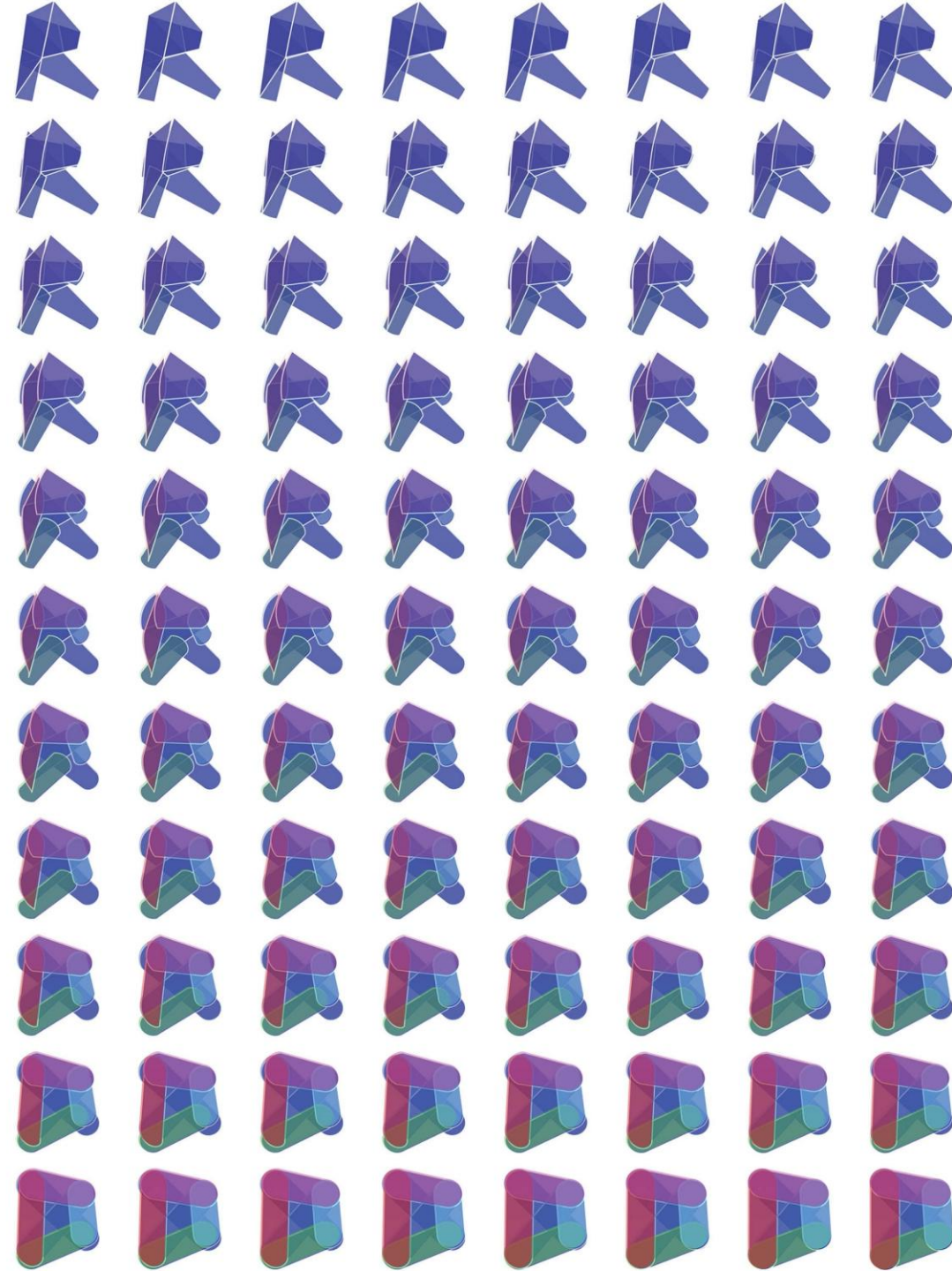


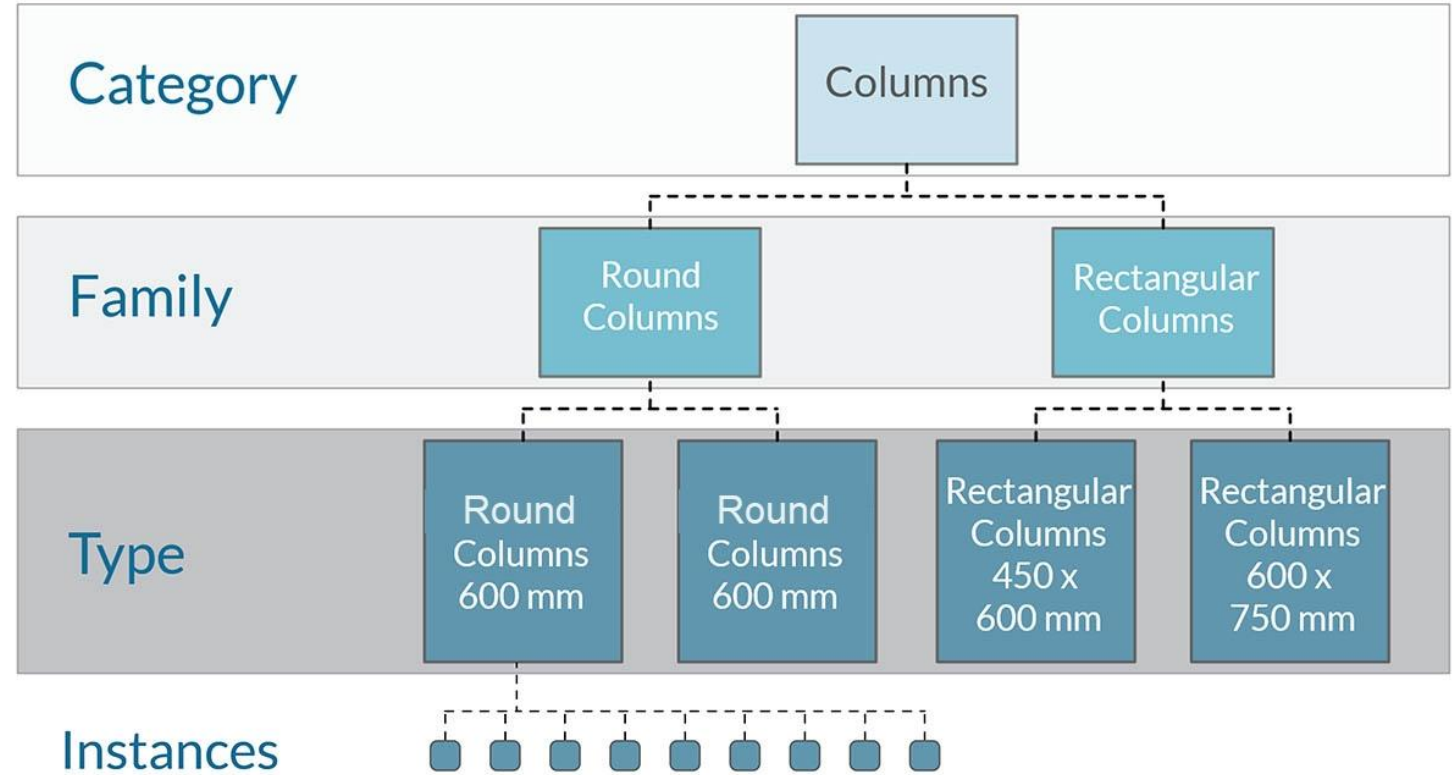
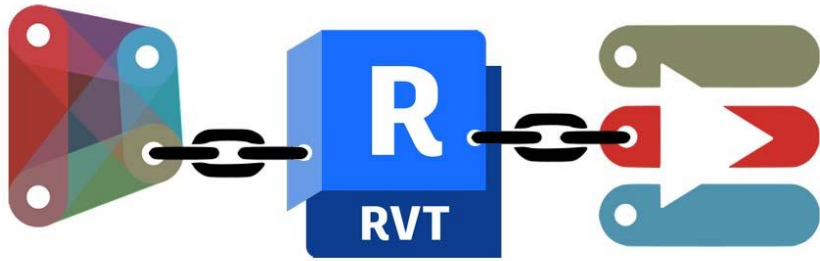
Point

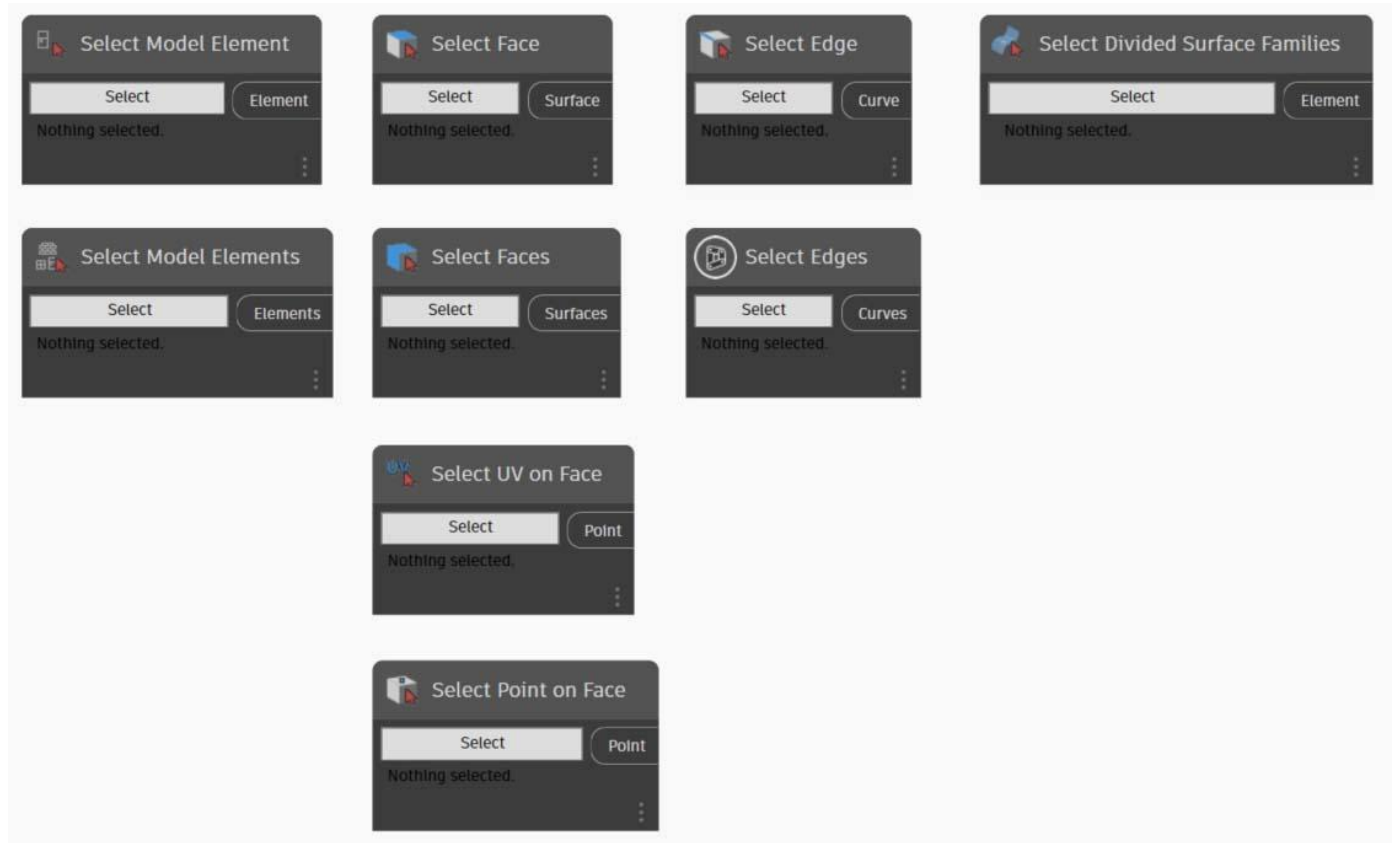
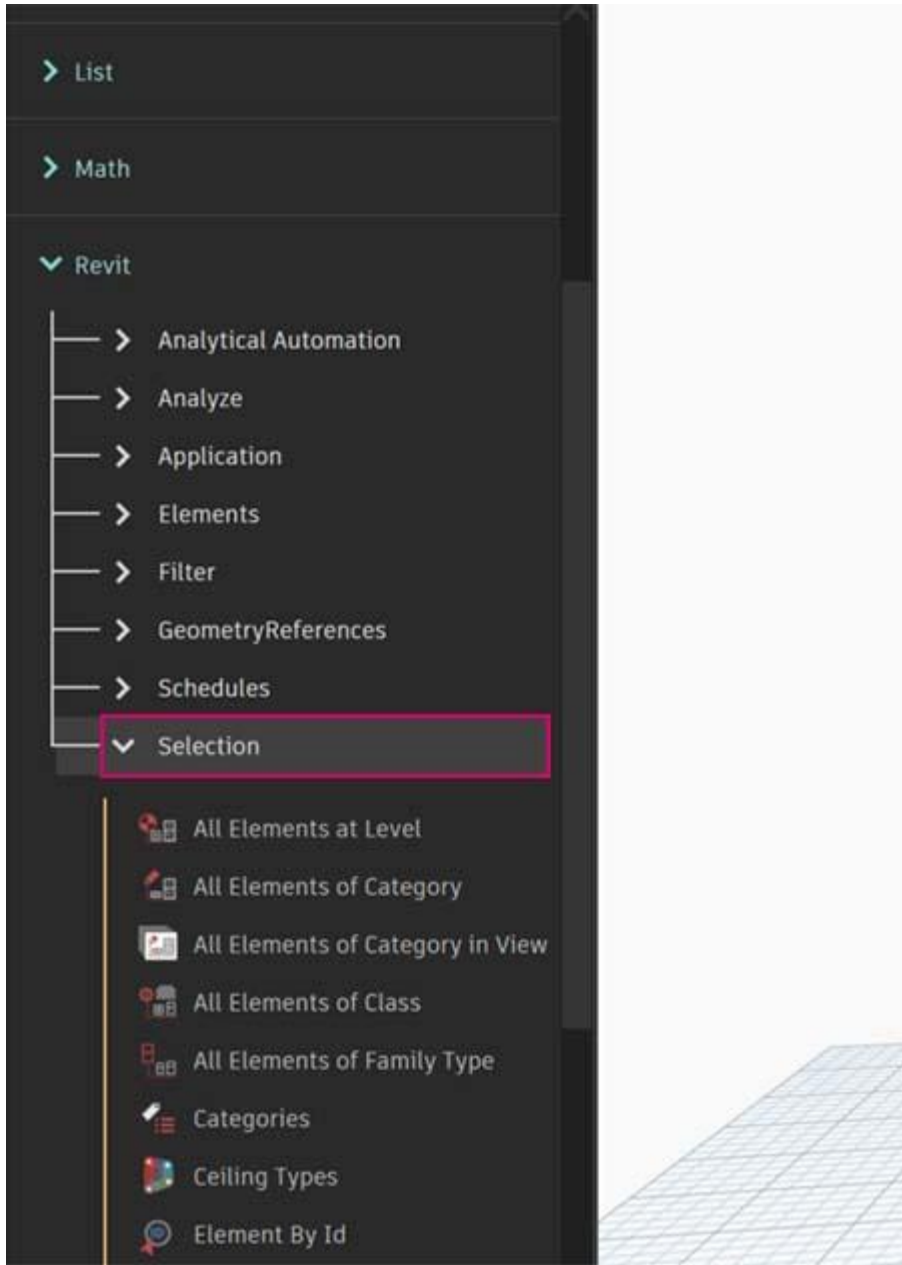


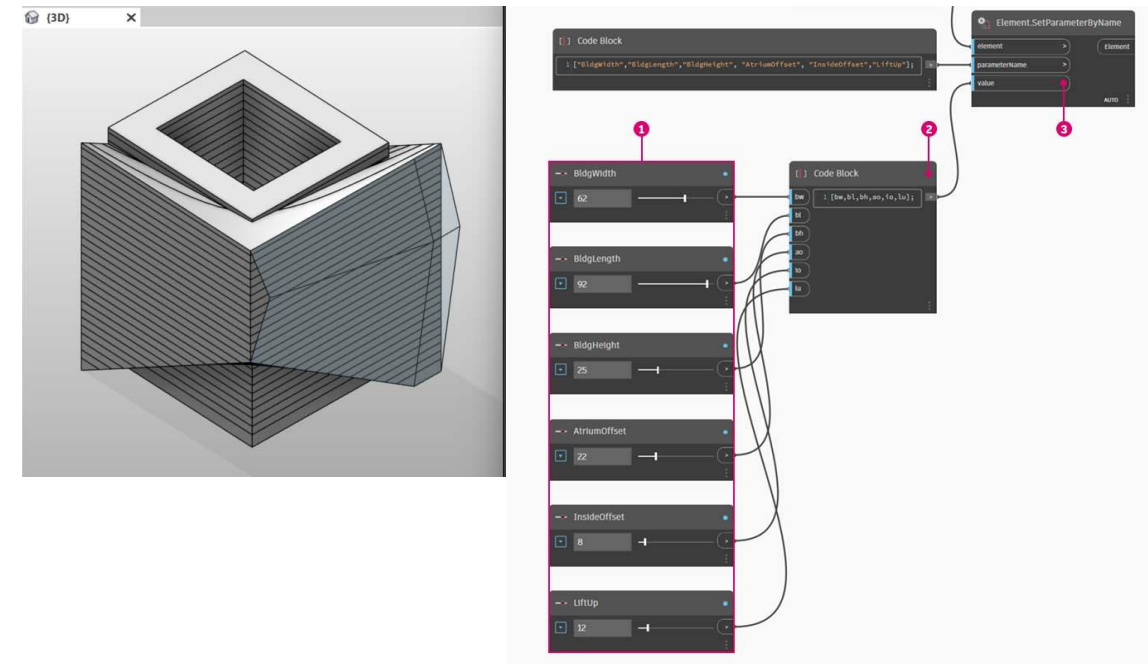
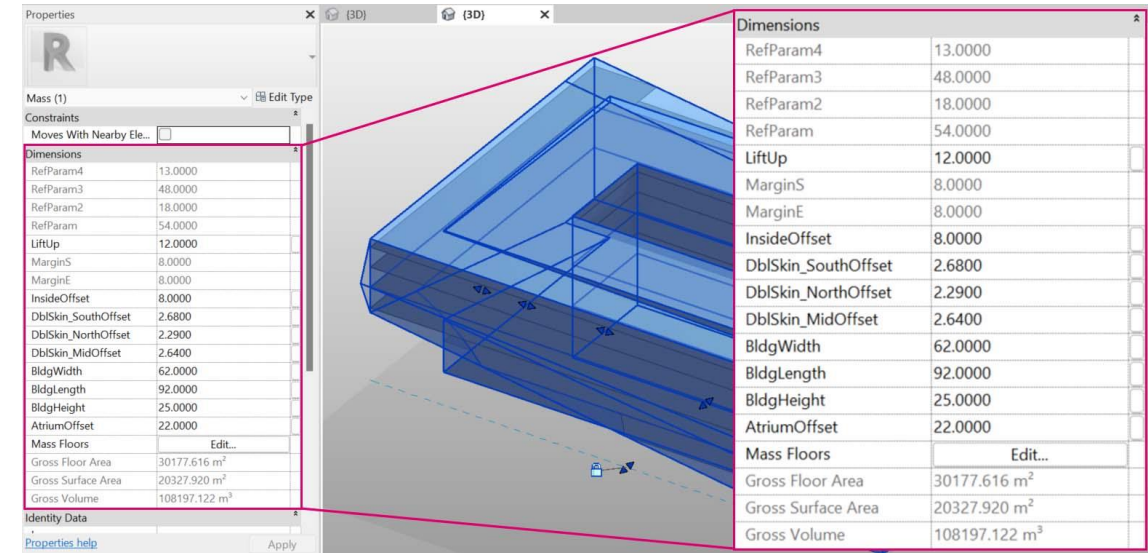
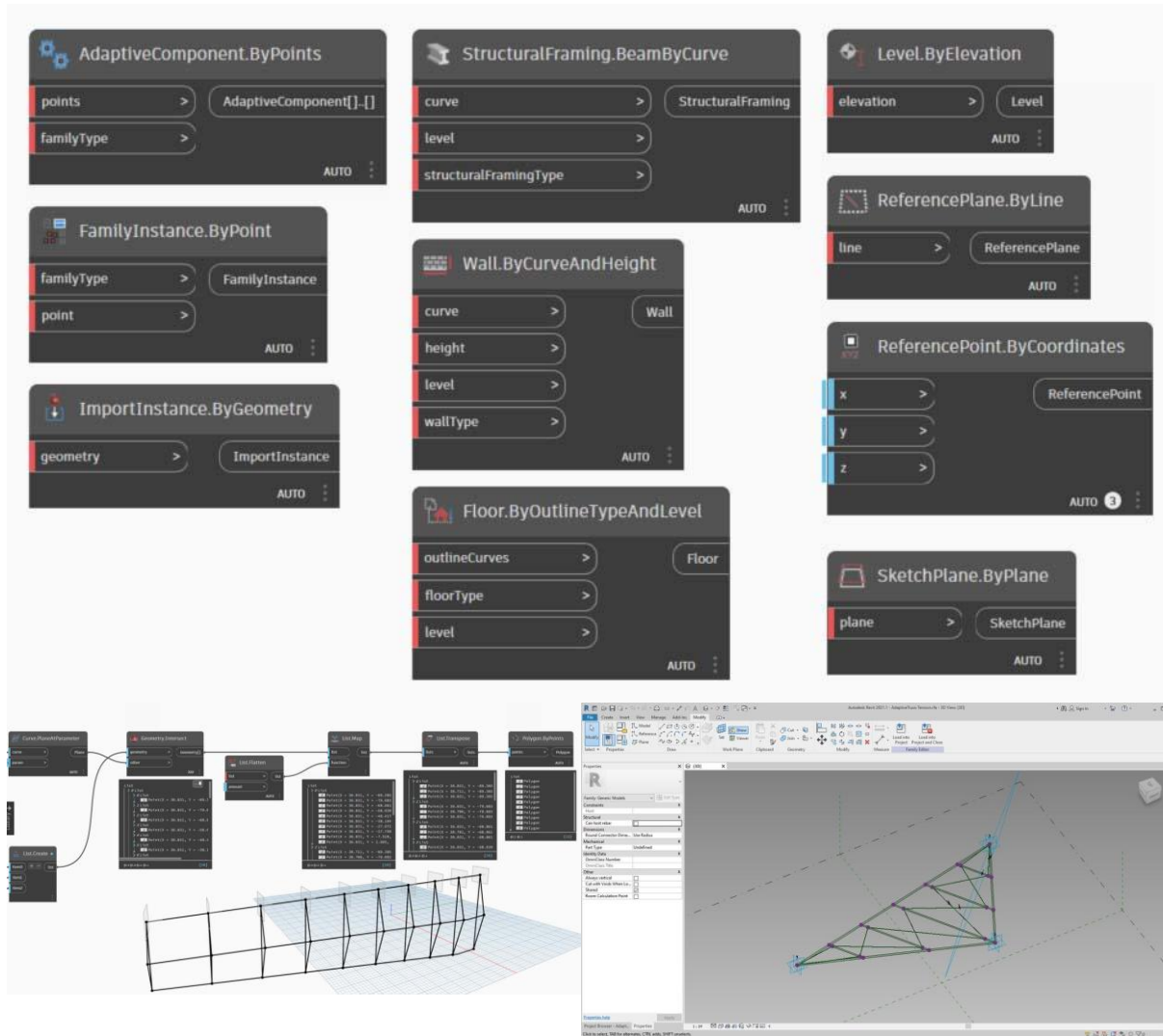
04 Geometry - Revit (Wall, Floor)

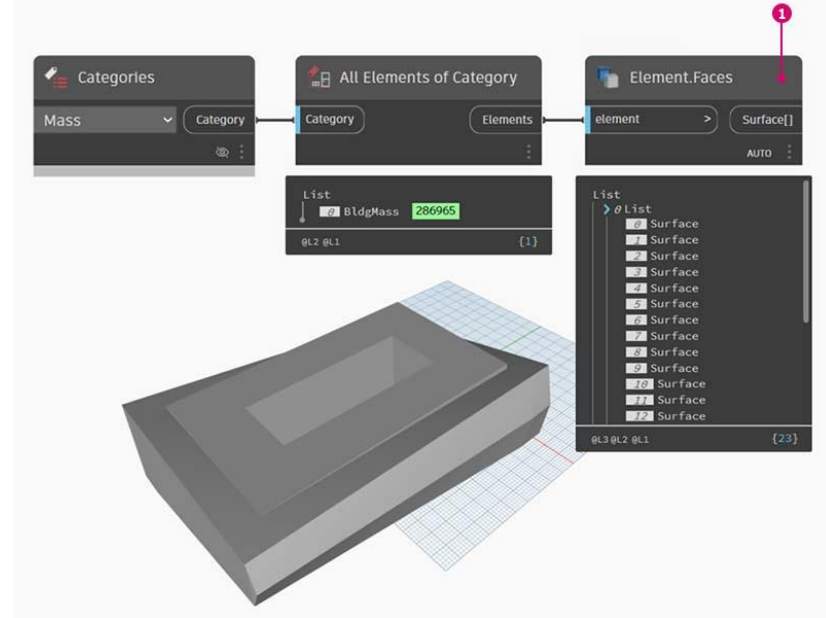
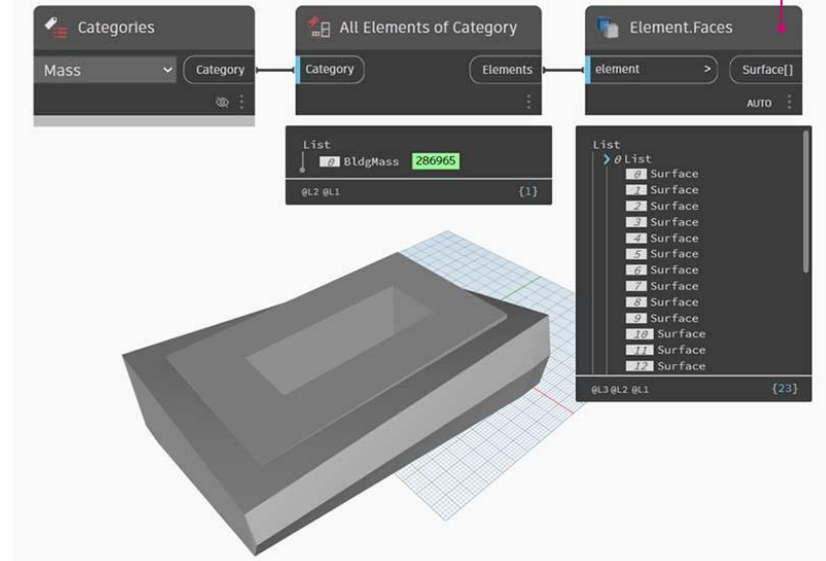
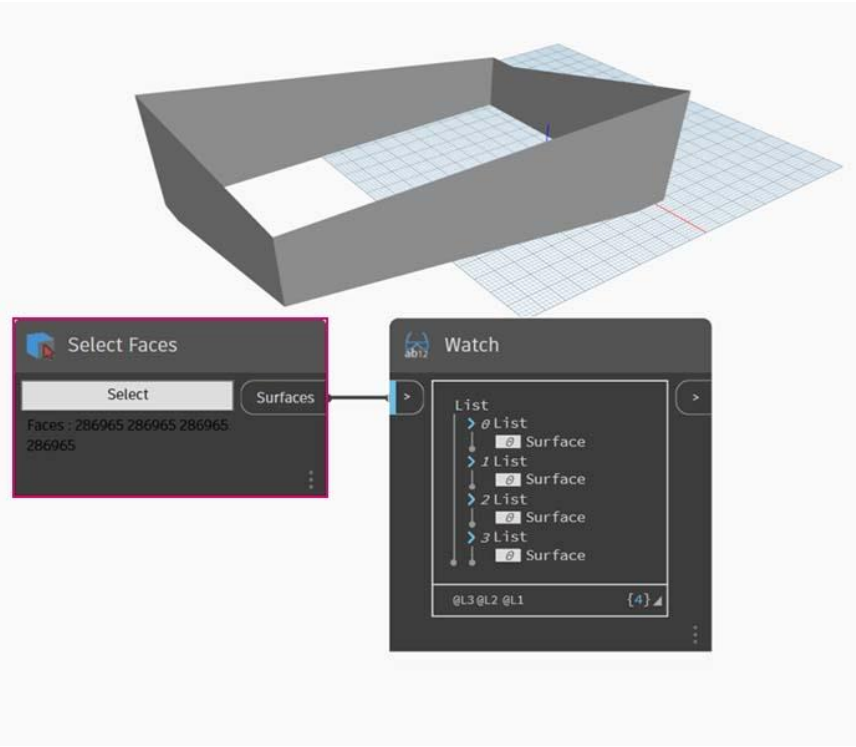
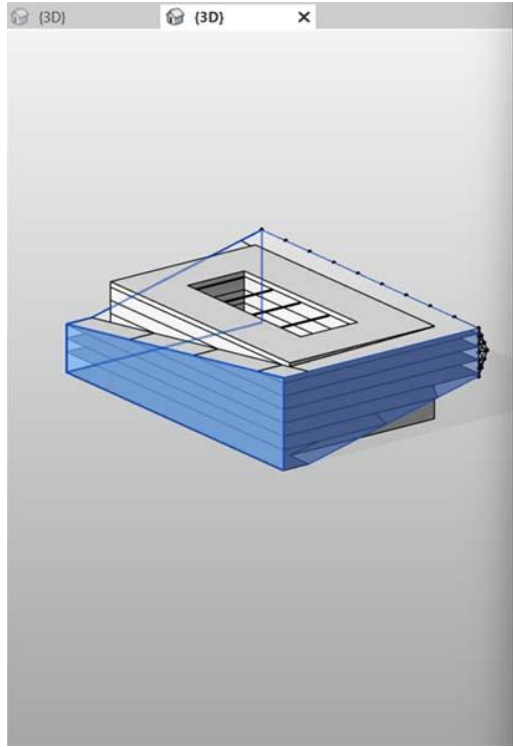
응용 및 Revit 정보 및 형상 연결

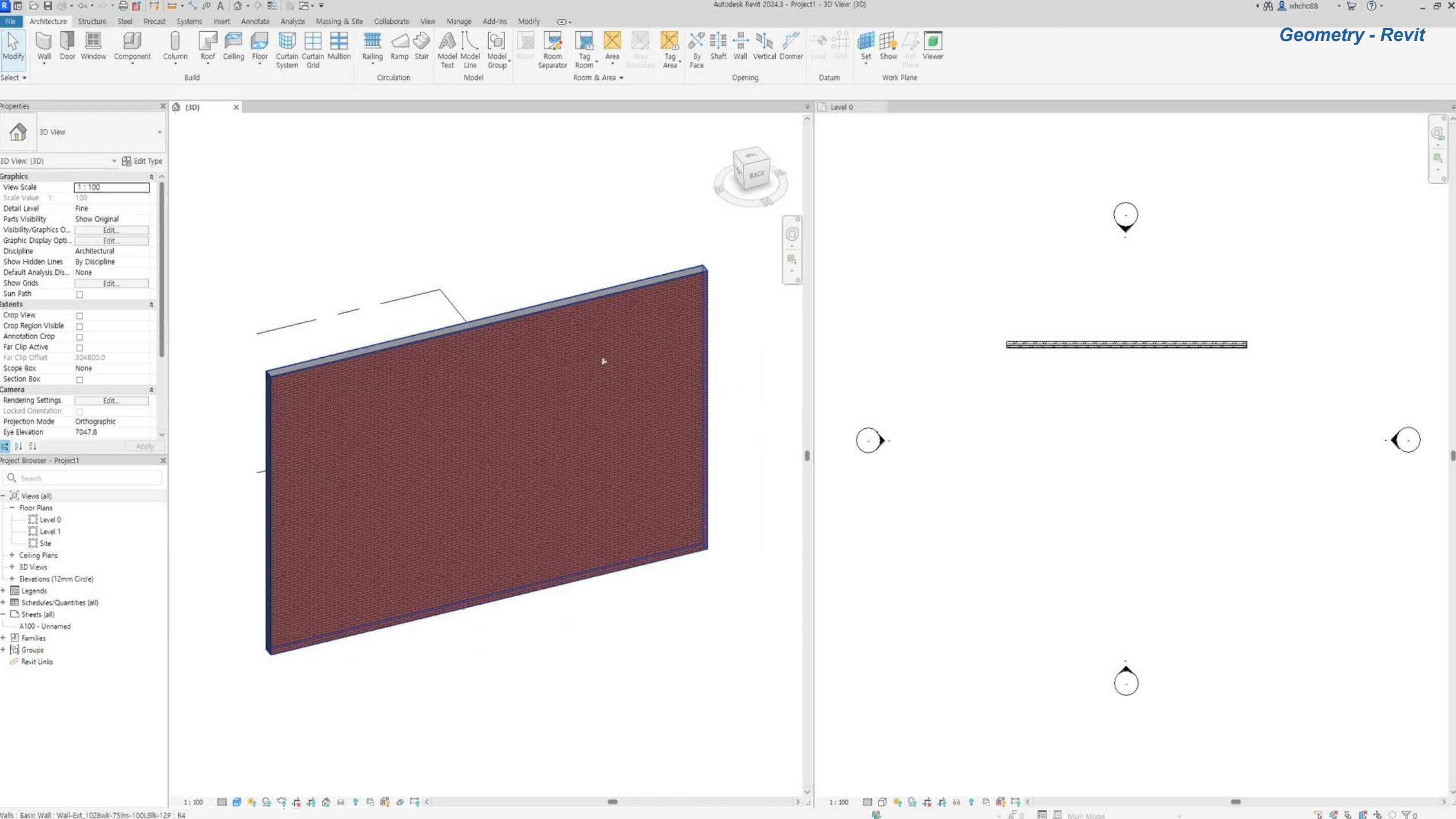


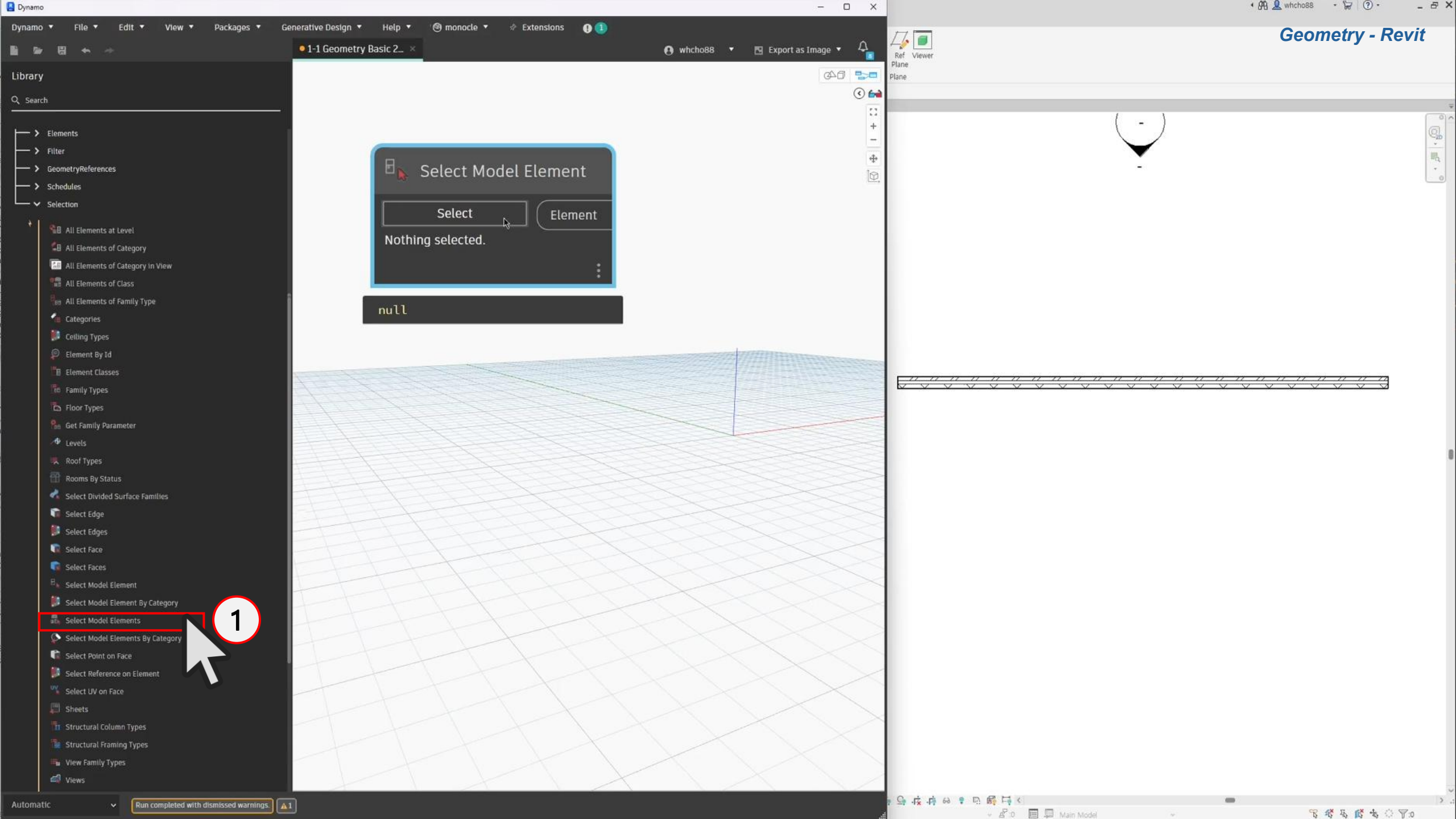


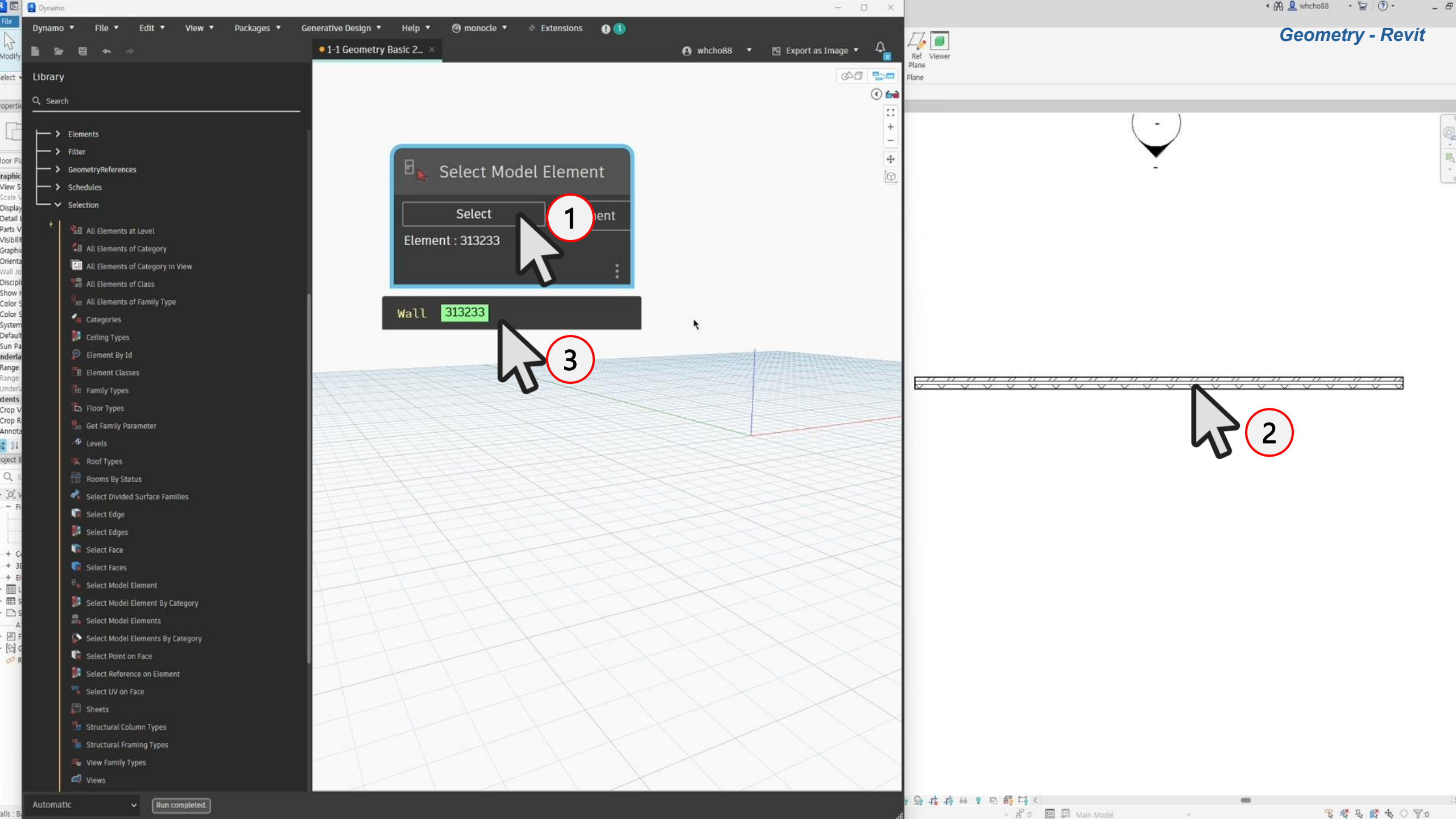














Select Model Element

Select

Element : 313233

Element

Wall 313233

Element.Geometry

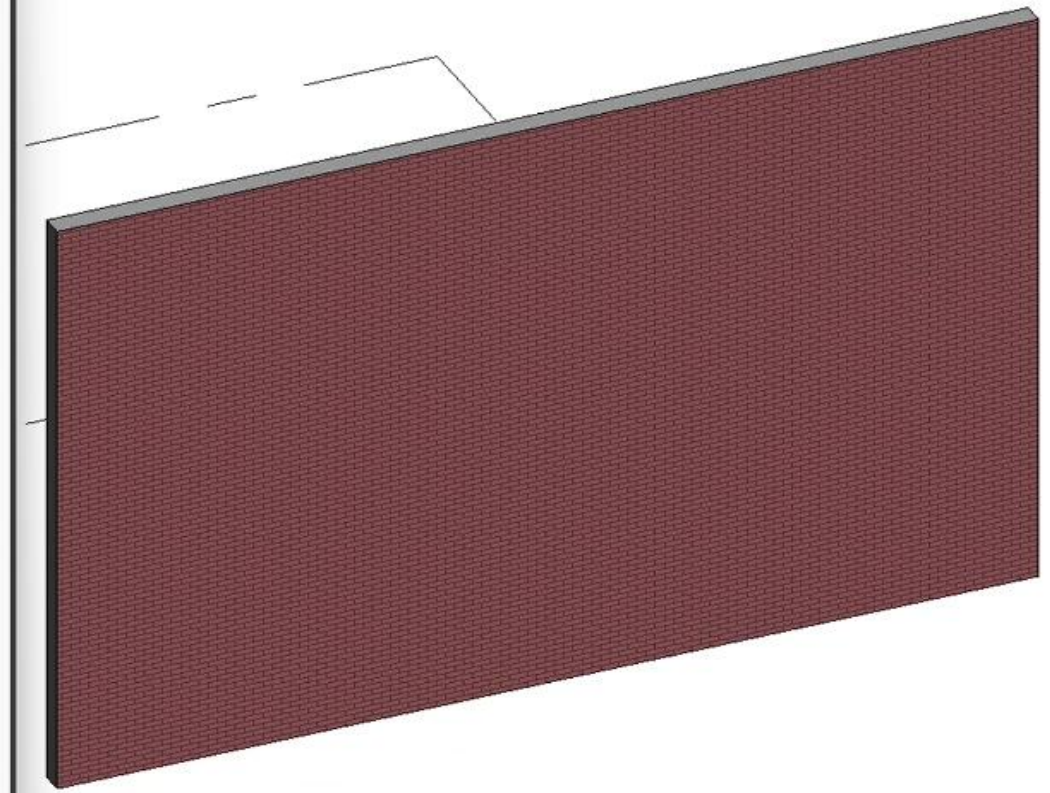
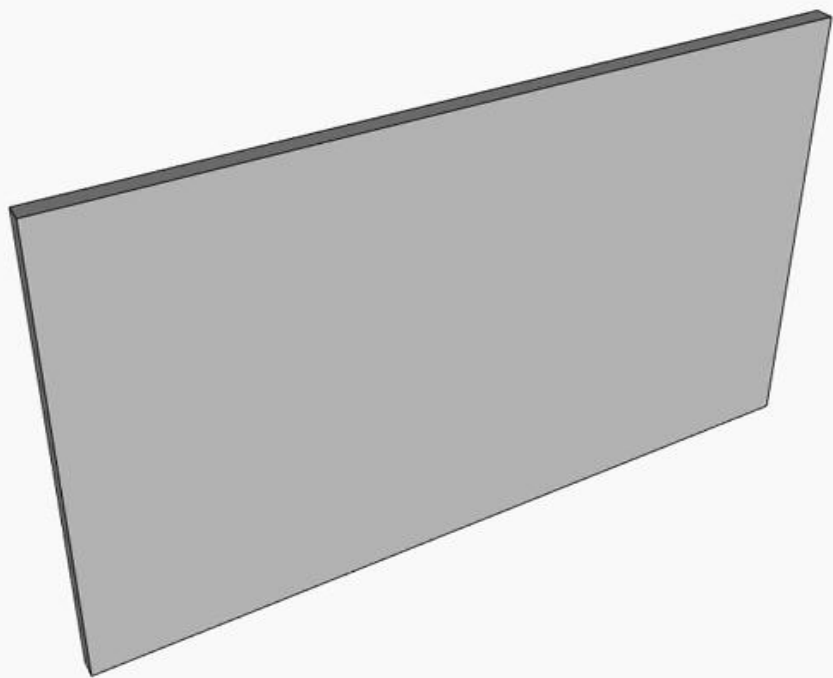
element > var[]

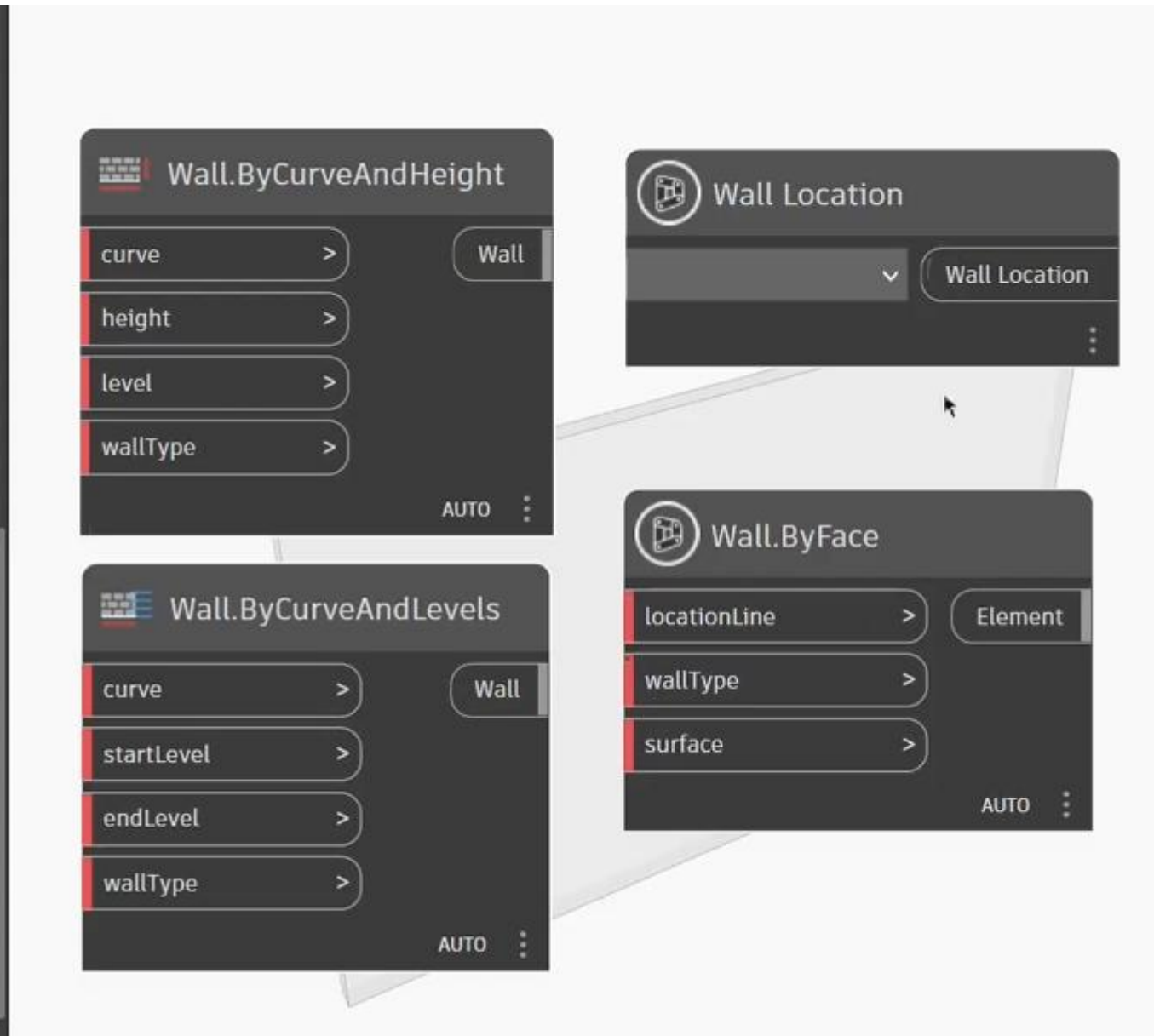
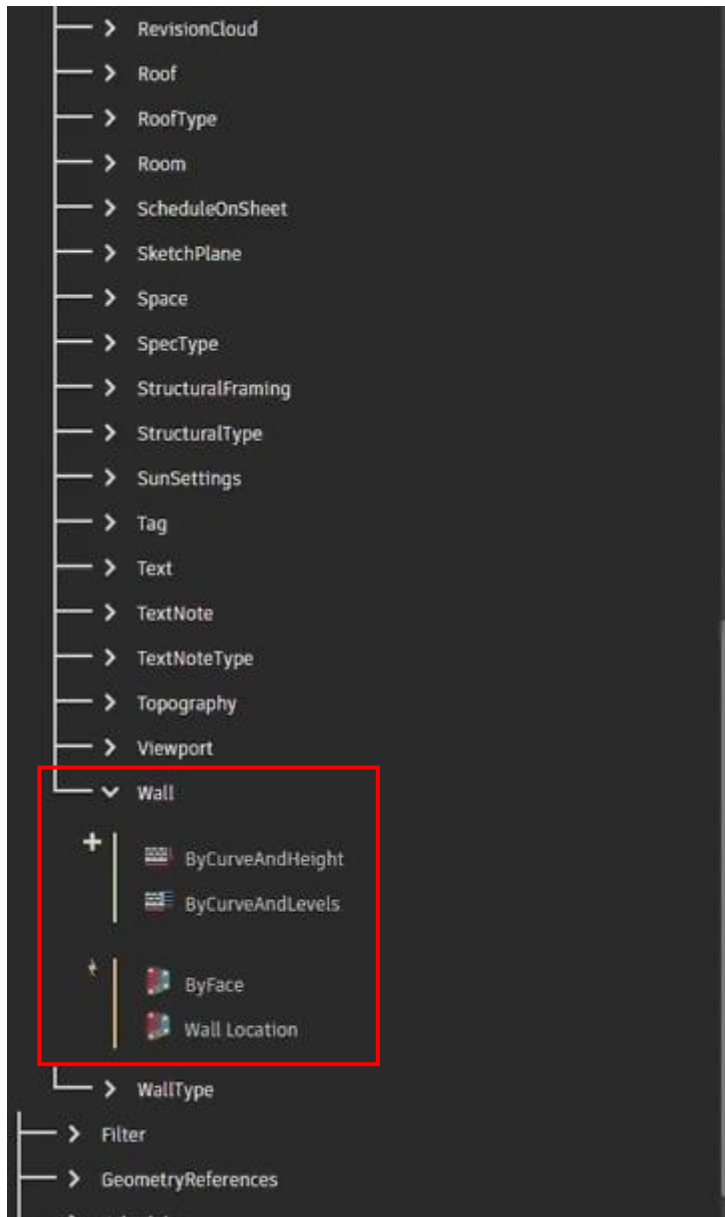
AUTO

List

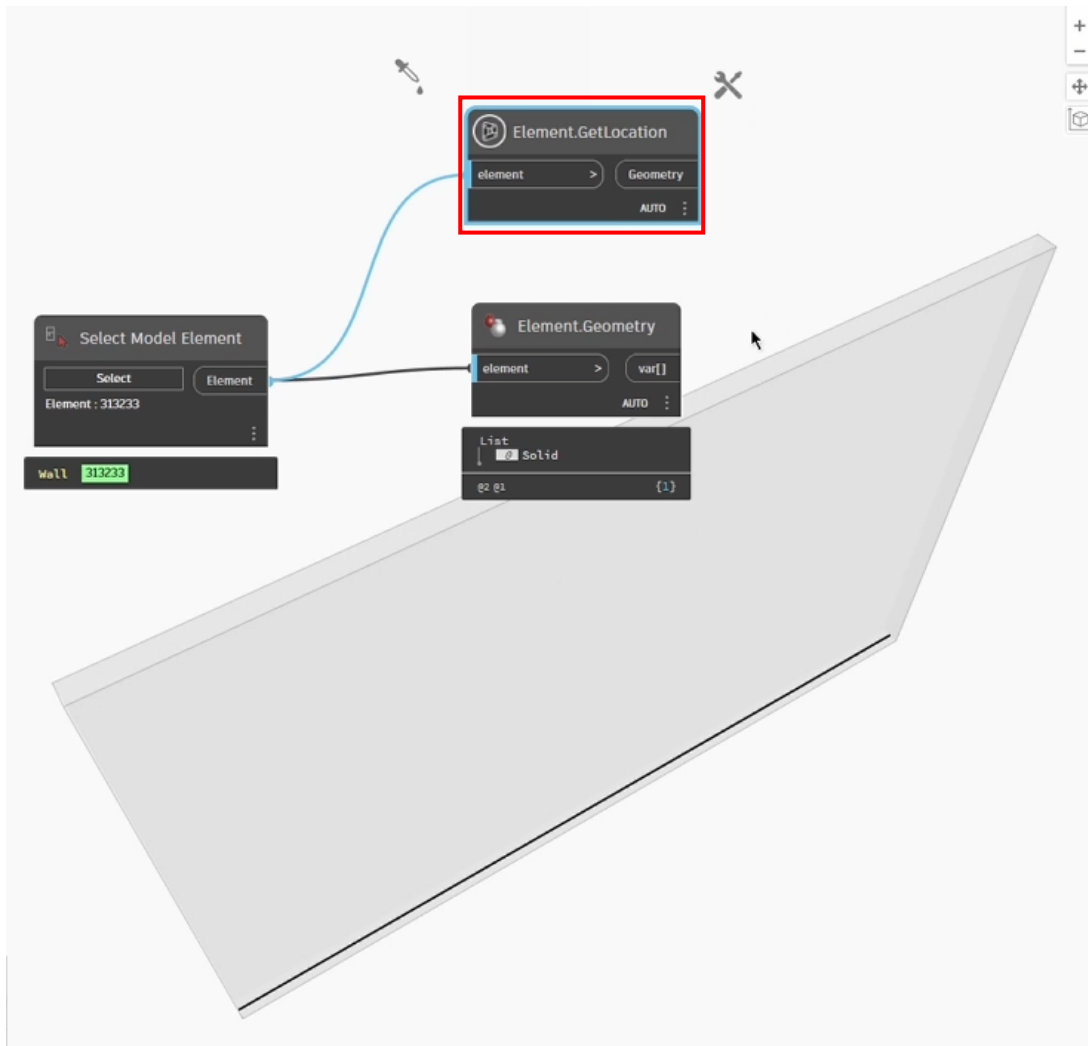
0 Solid

@2 @1 {1}

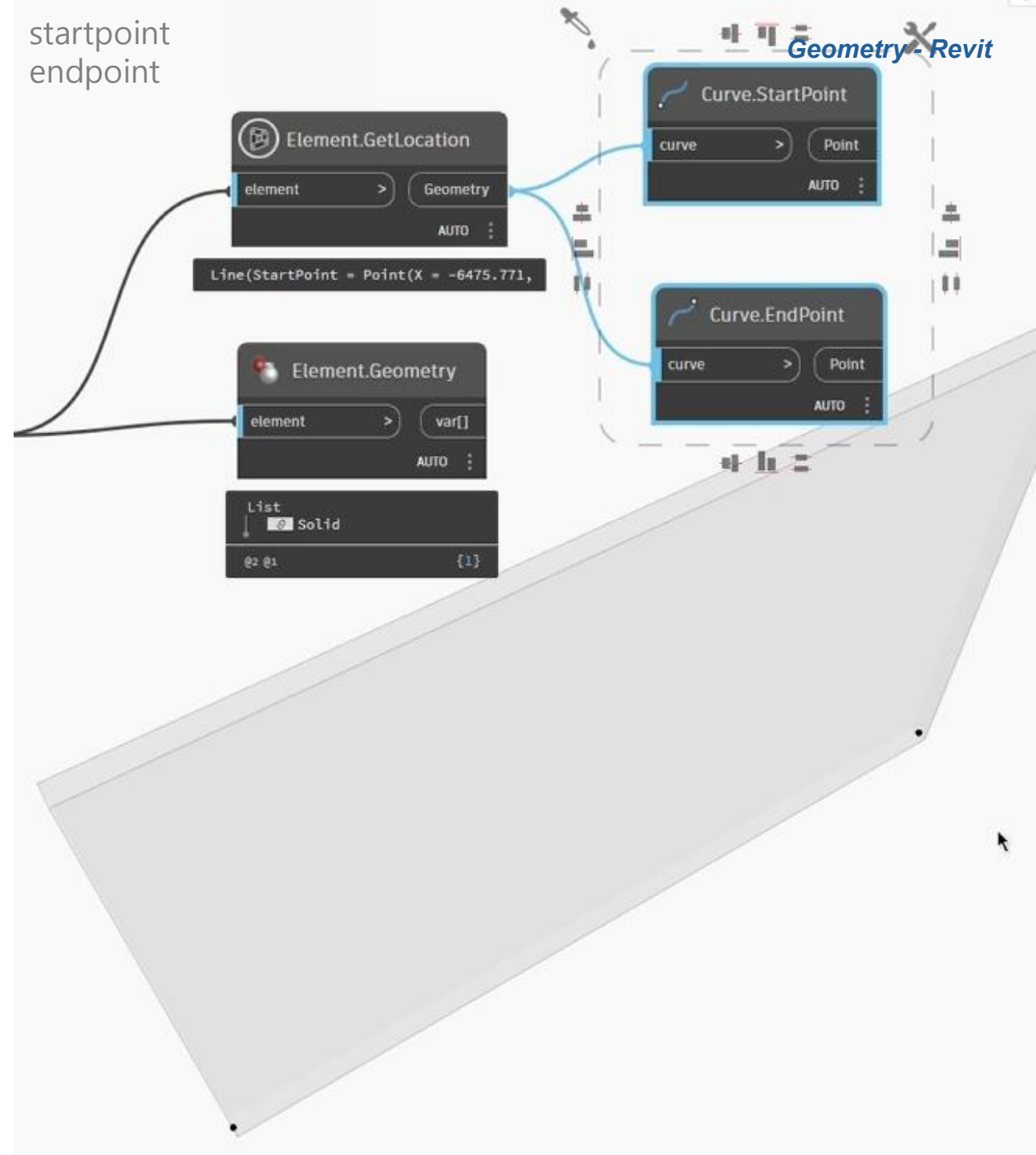




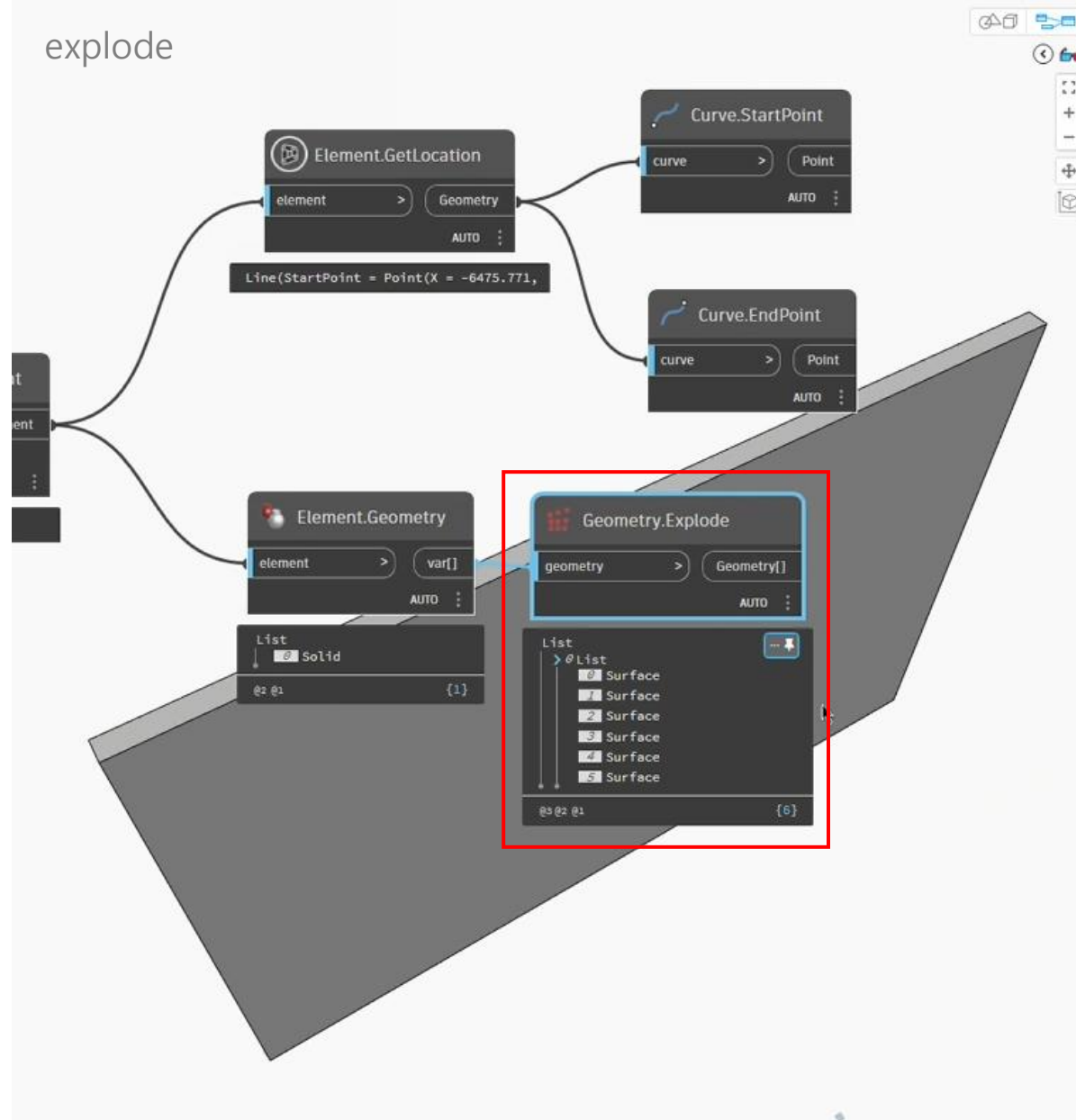
GetLocation



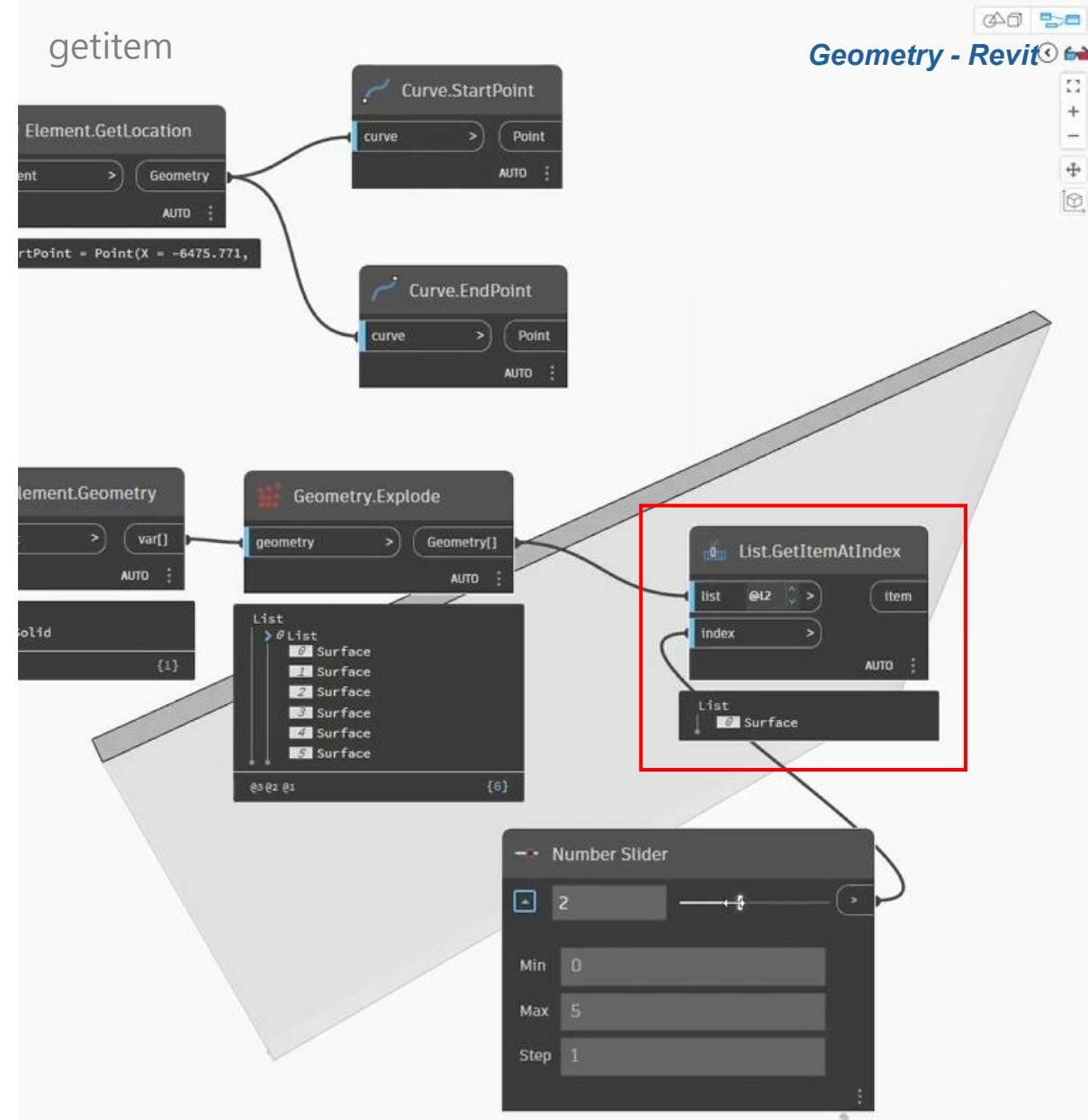
```
startpoint
endpoint
```



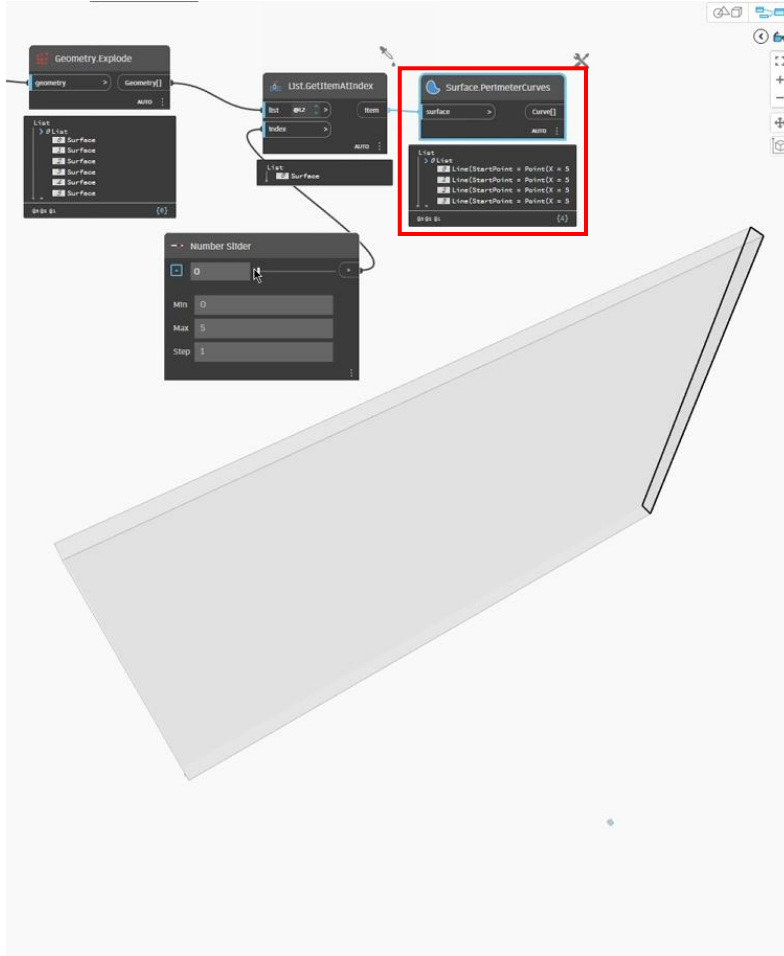
explode



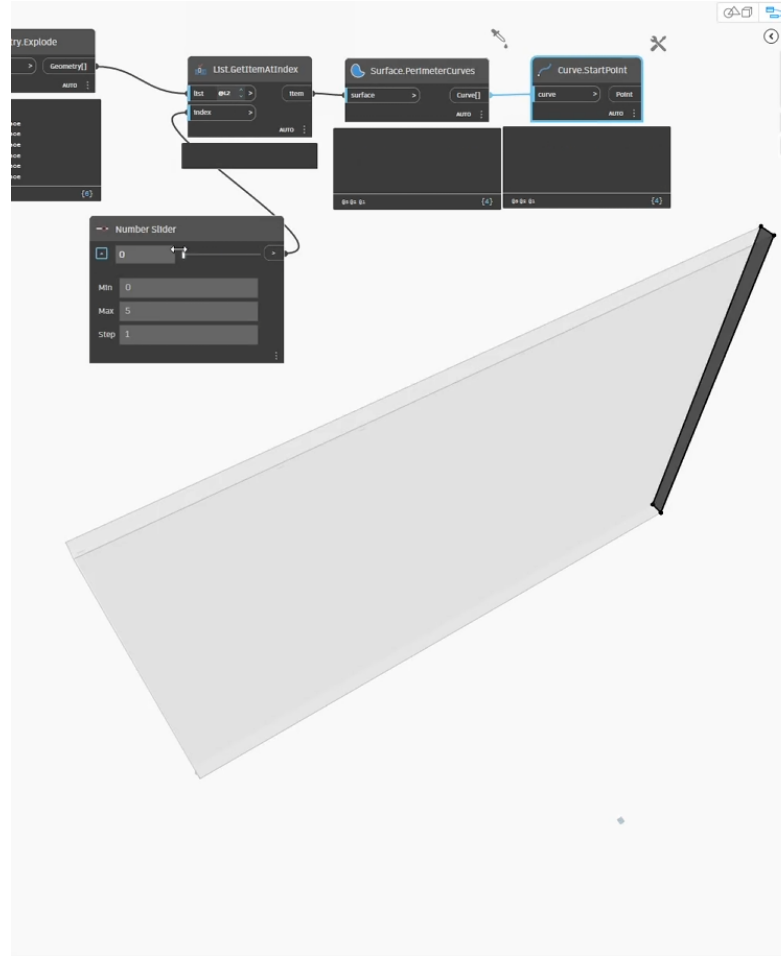
getitem



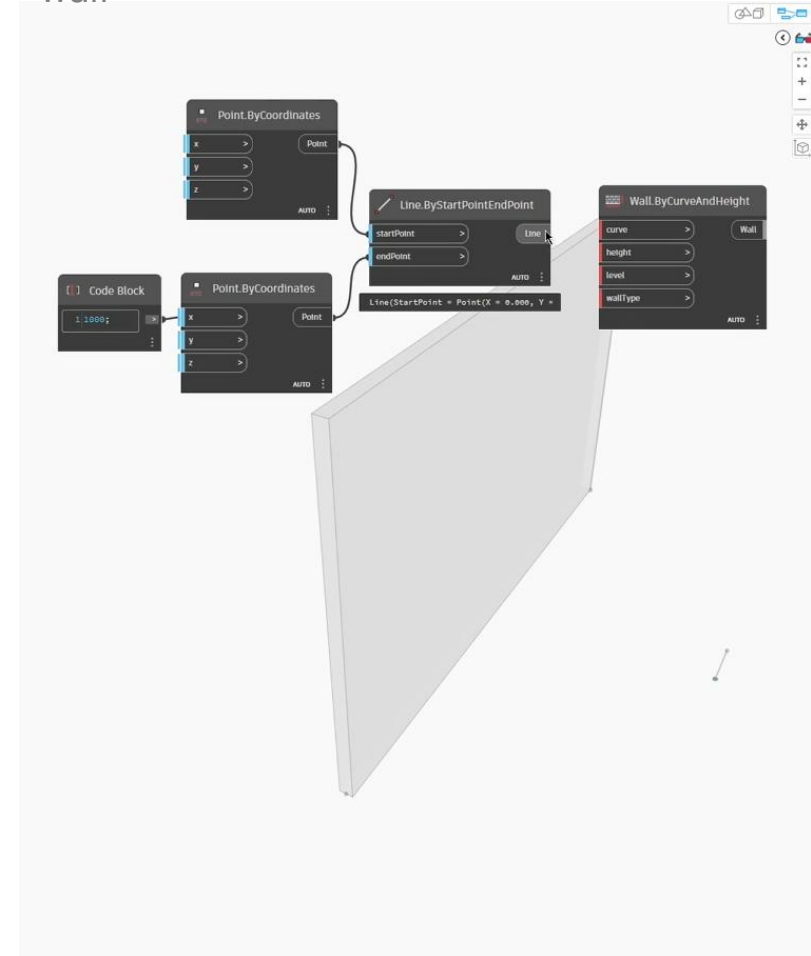
perimeter

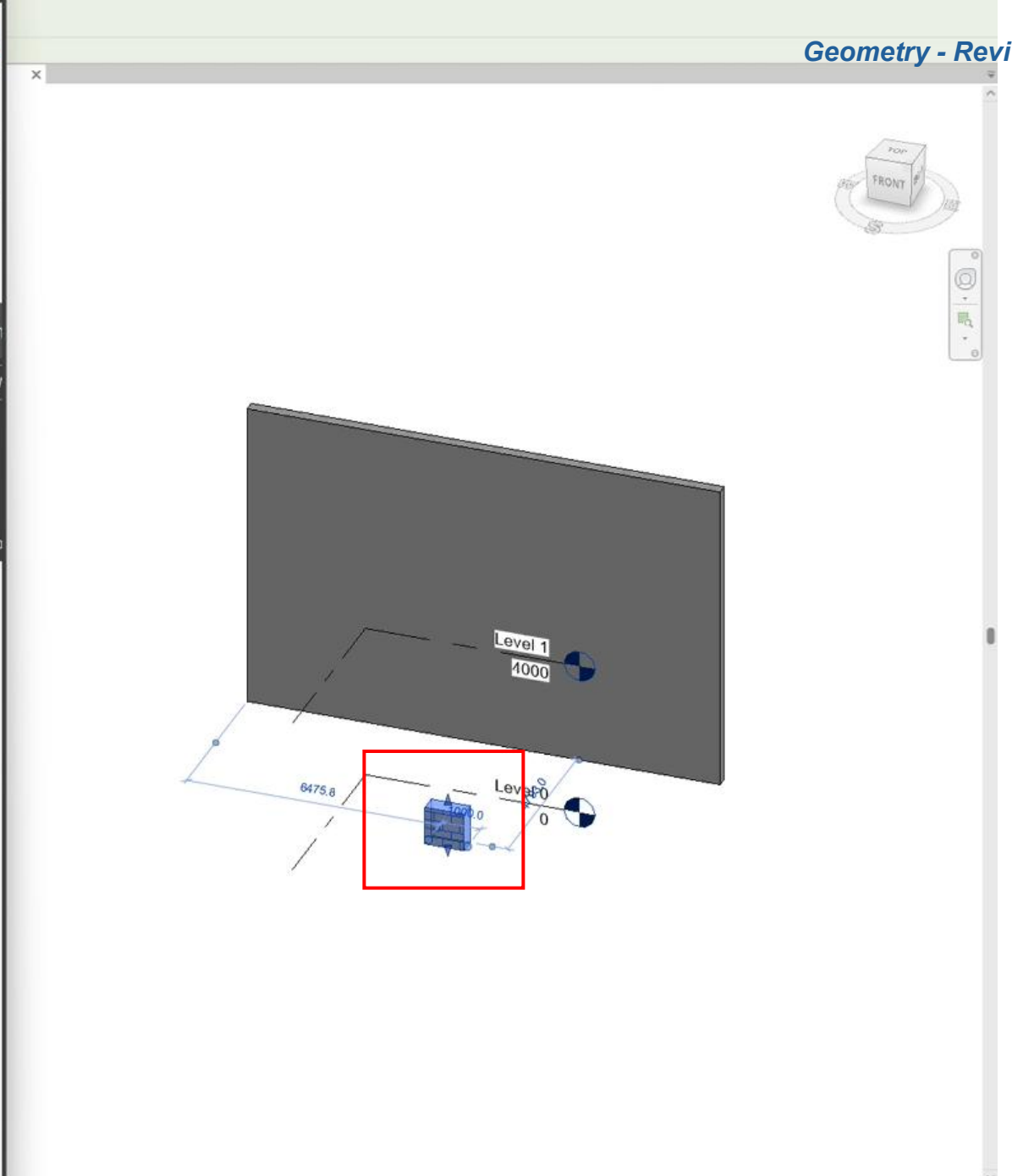
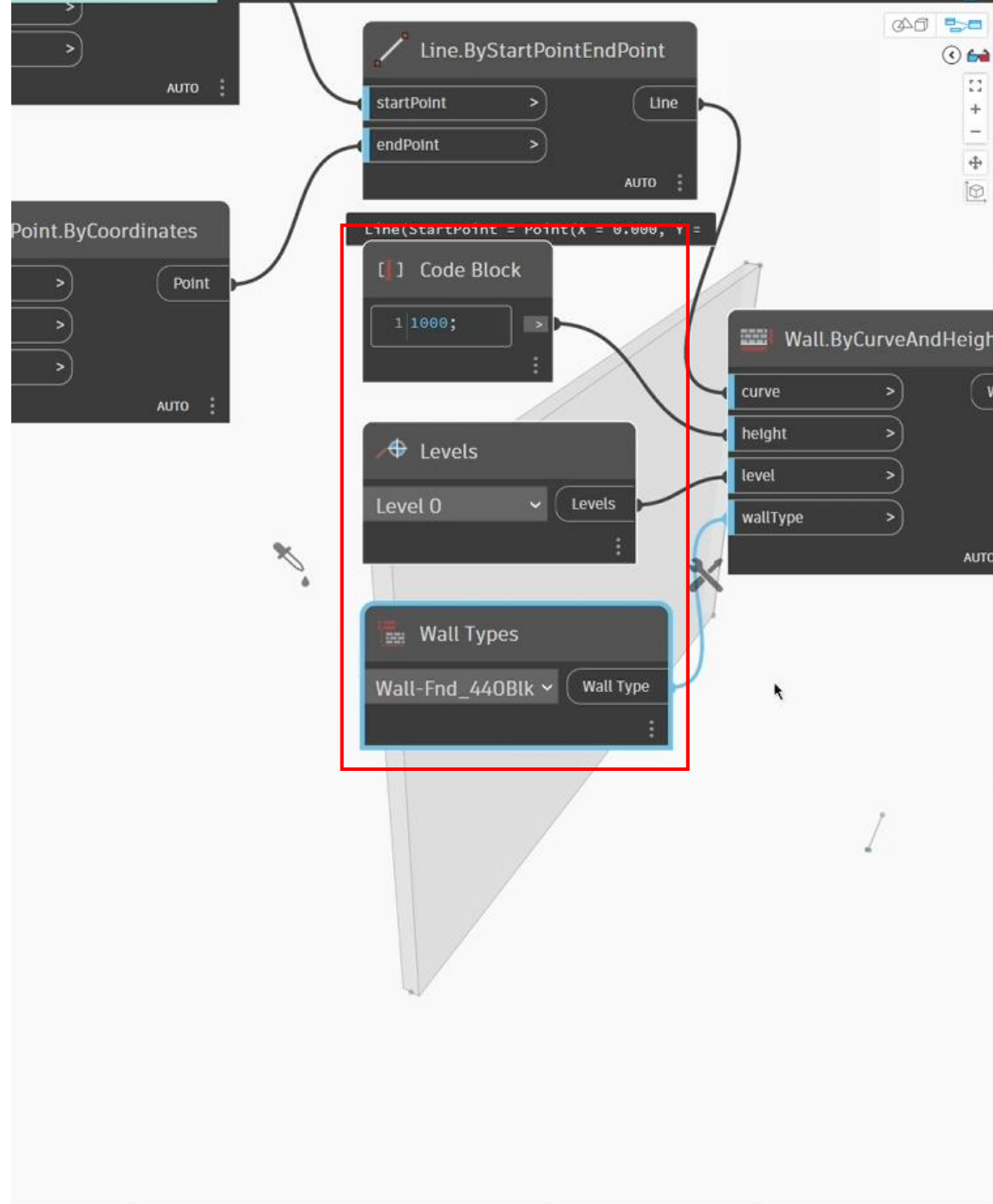


startpoint

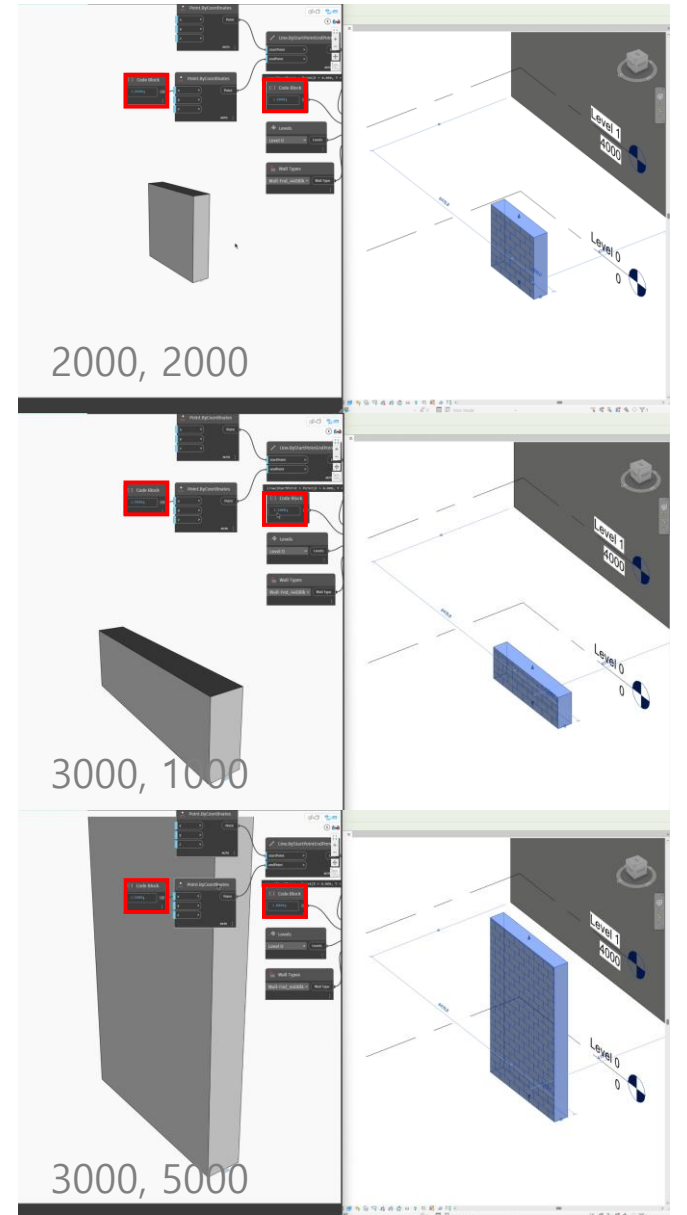
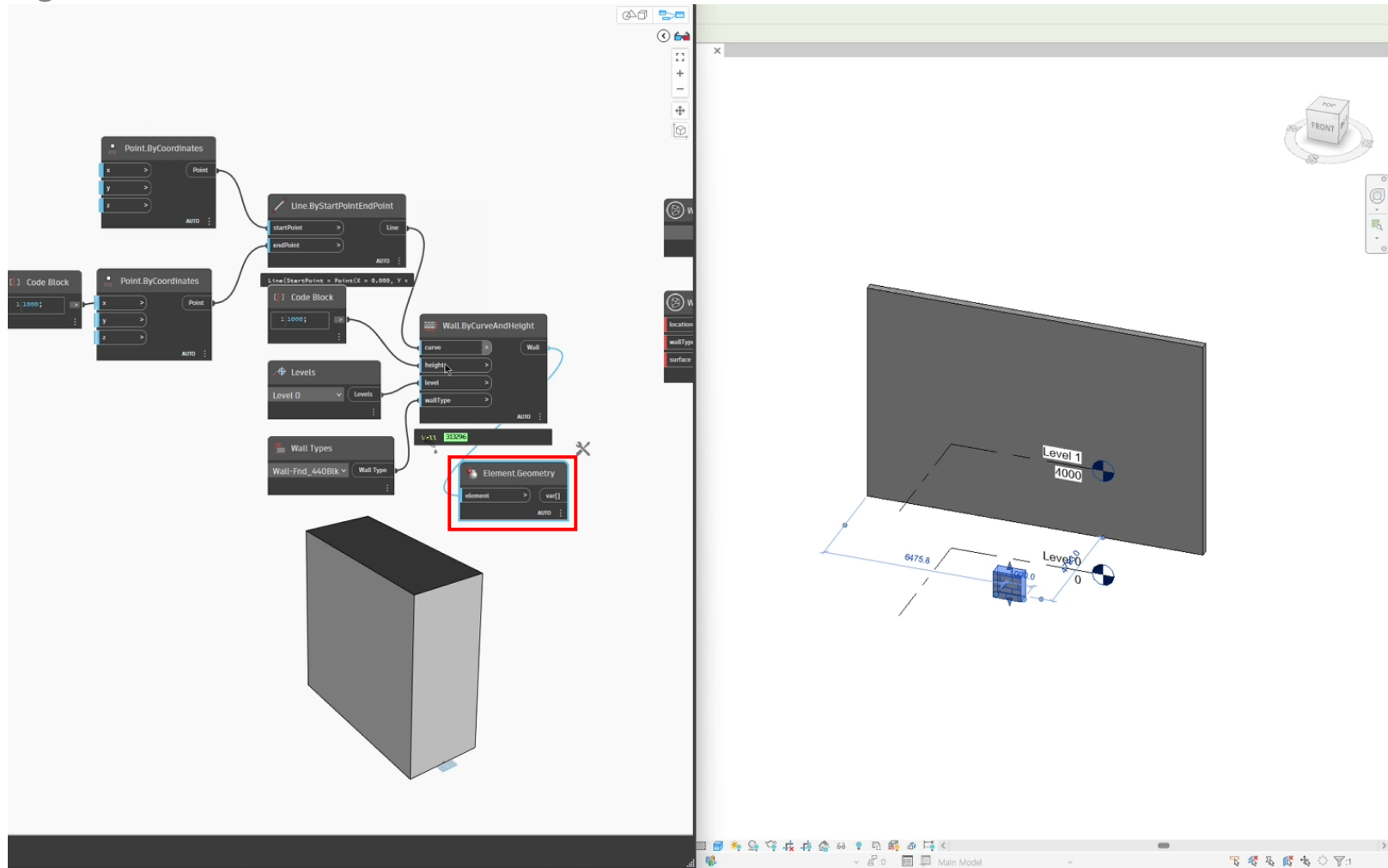


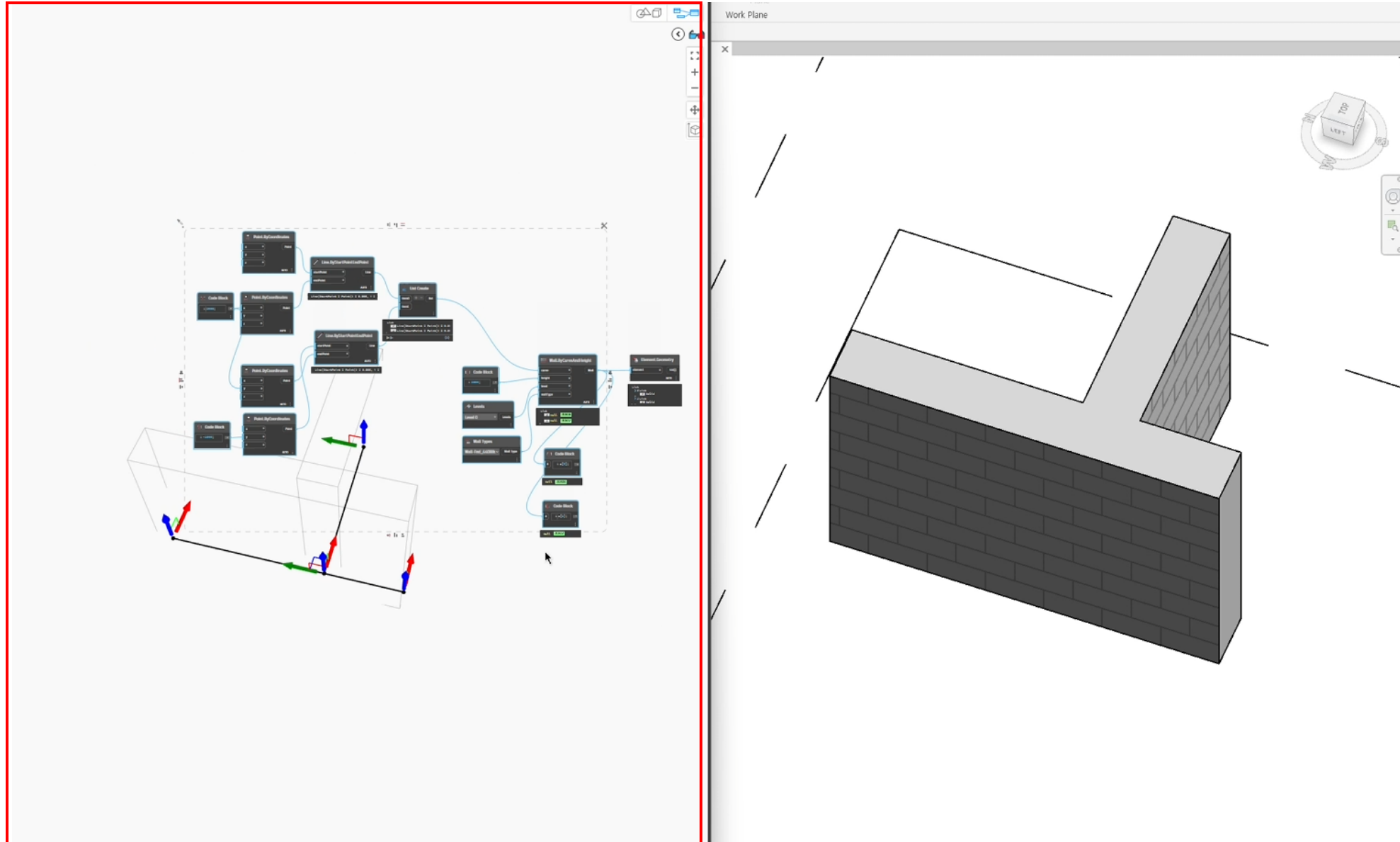
Line
wall

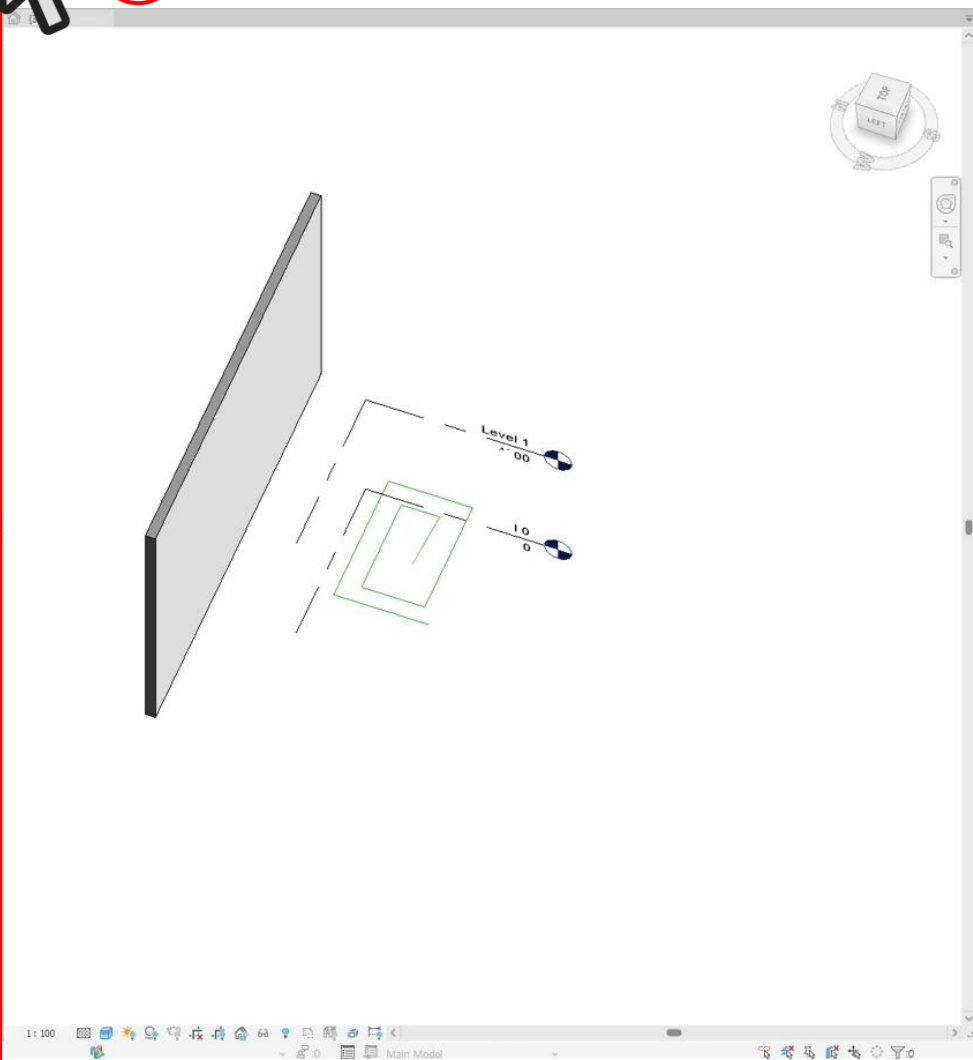
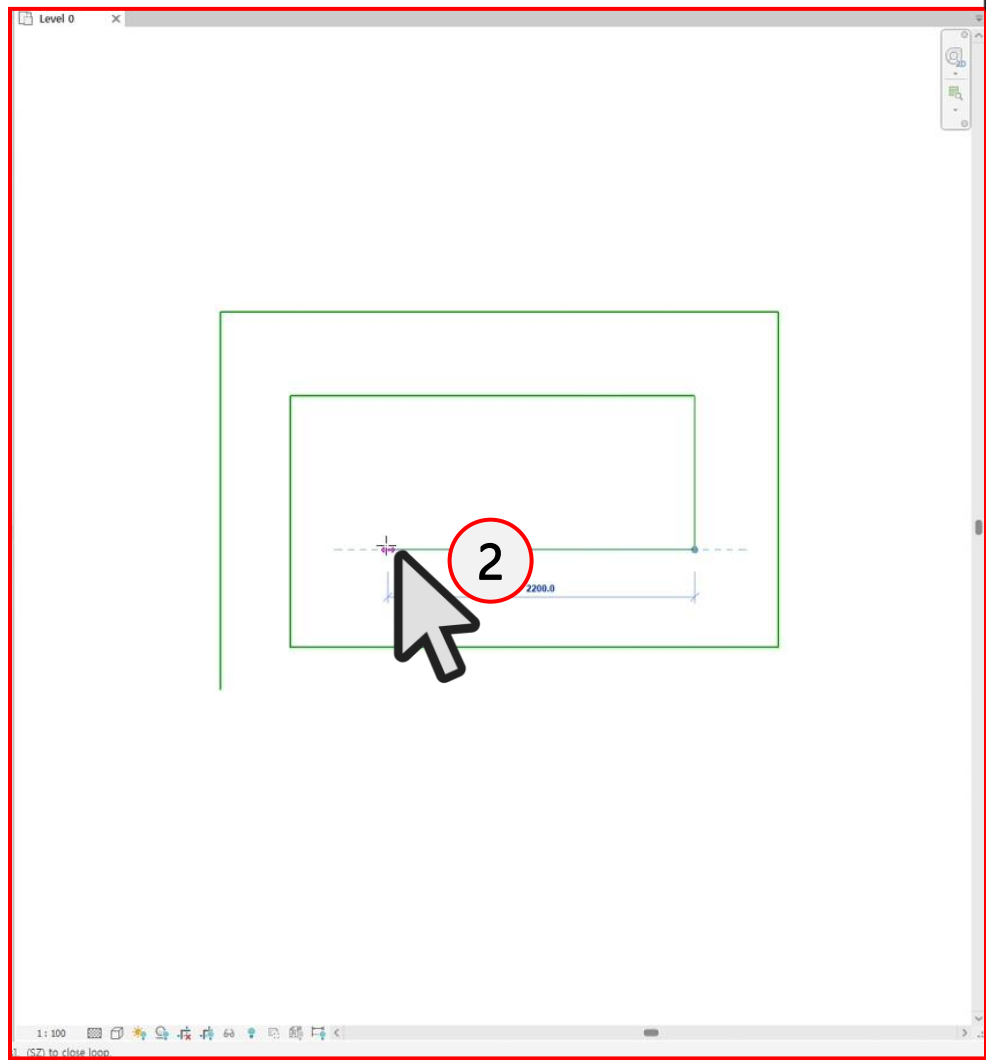
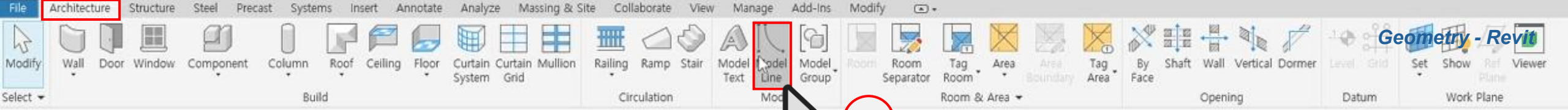




geo







Select model elements
geo

Select Model Elements

Select

Elements

Elements : 313572 313717 313821
313989 314074 314171 314252
314349

0	ModelCurve	313572
1	ModelCurve	313717
2	ModelCurve	313821
3	ModelCurve	313989
4	ModelCurve	314074
5	ModelCurve	314171
6	ModelCurve	314252
7	ModelCurve	314349

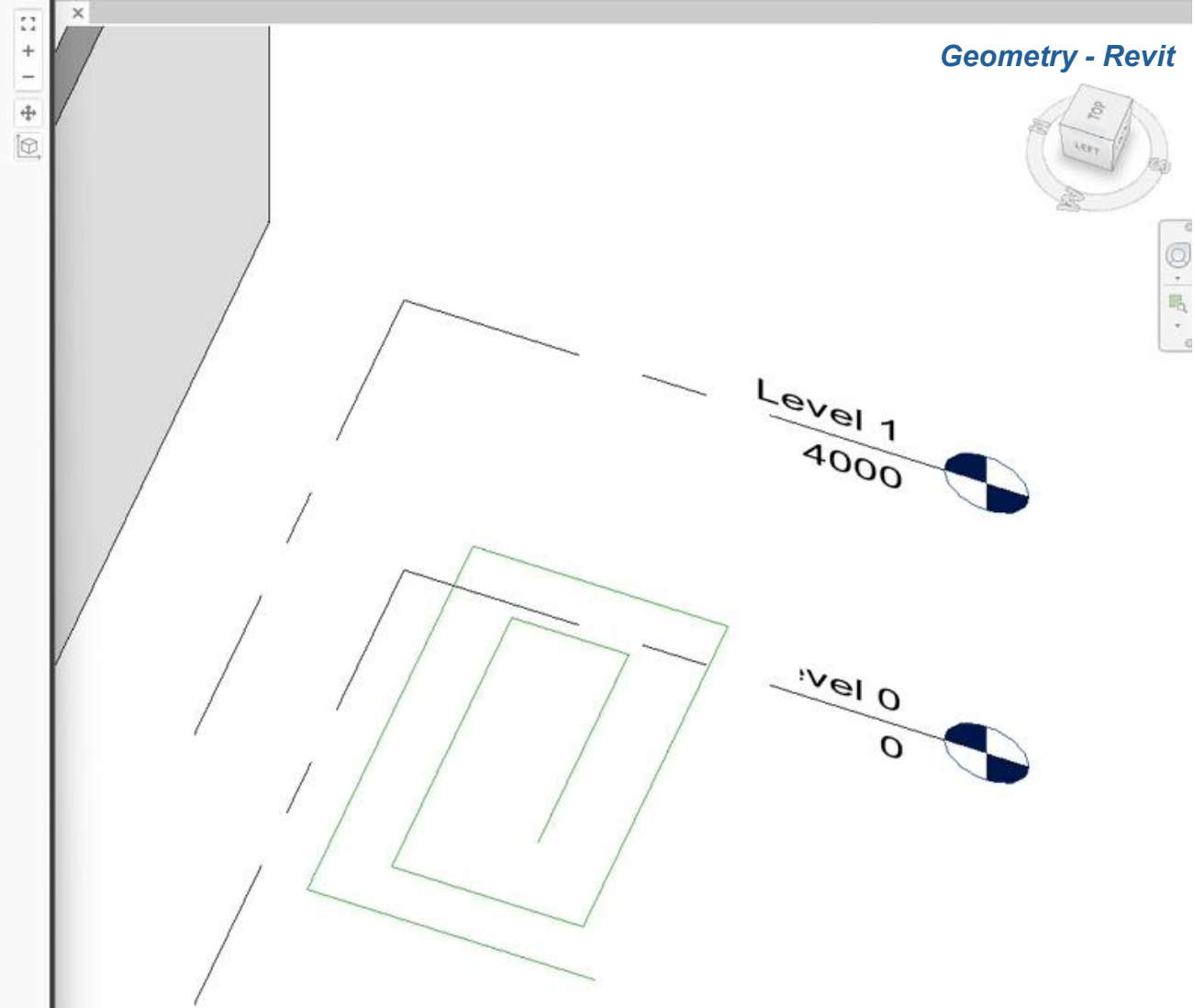
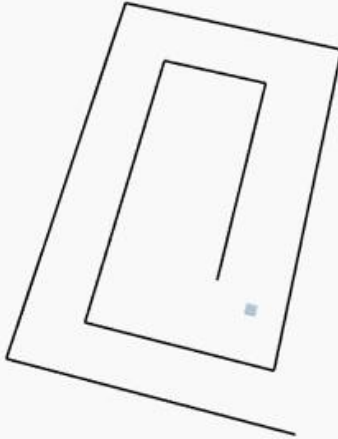
@2 @1 {8}

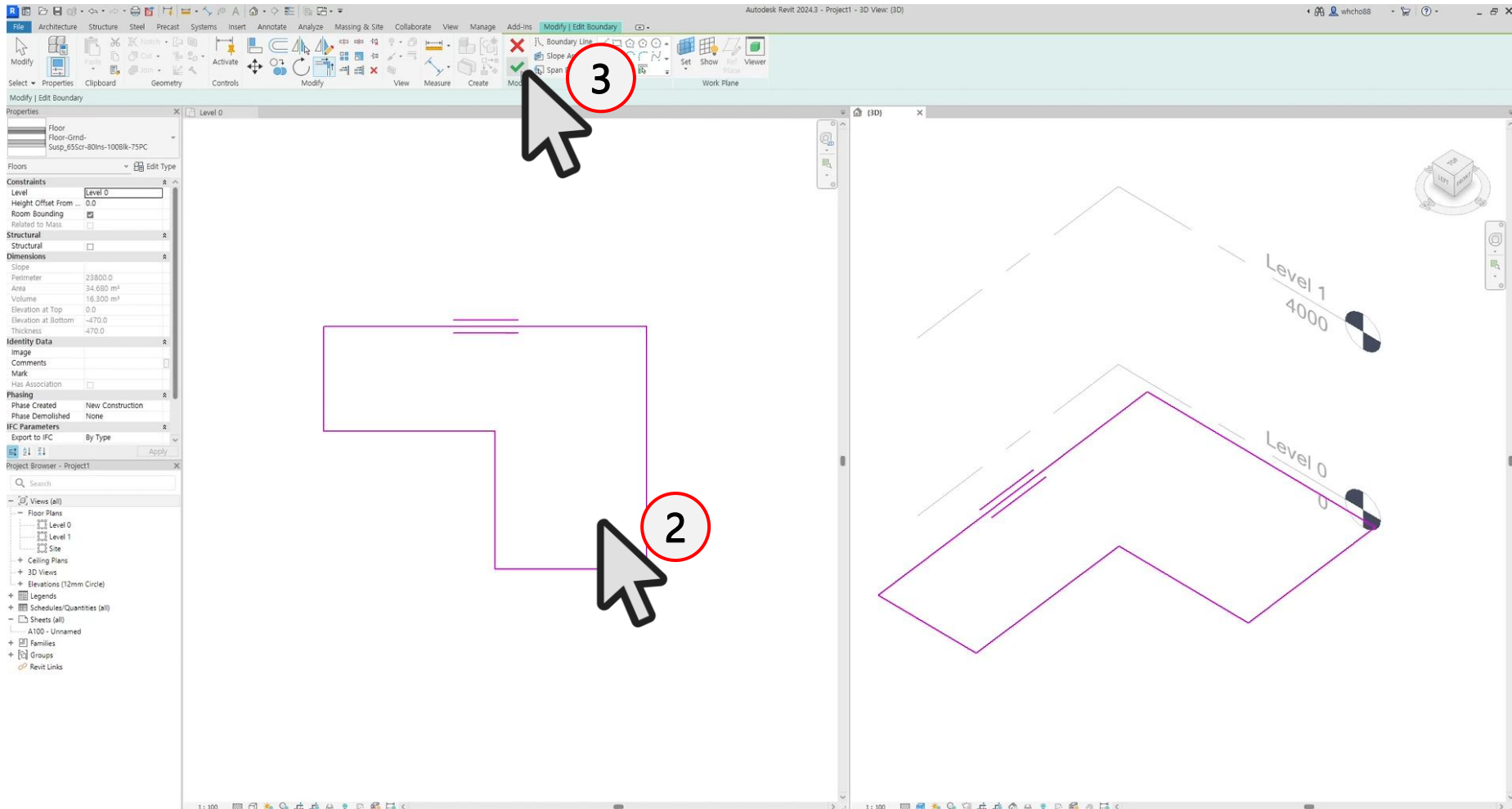
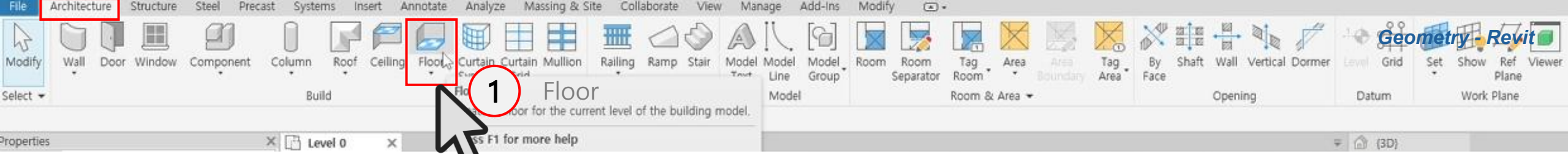
Element.Geometry

element > var[]

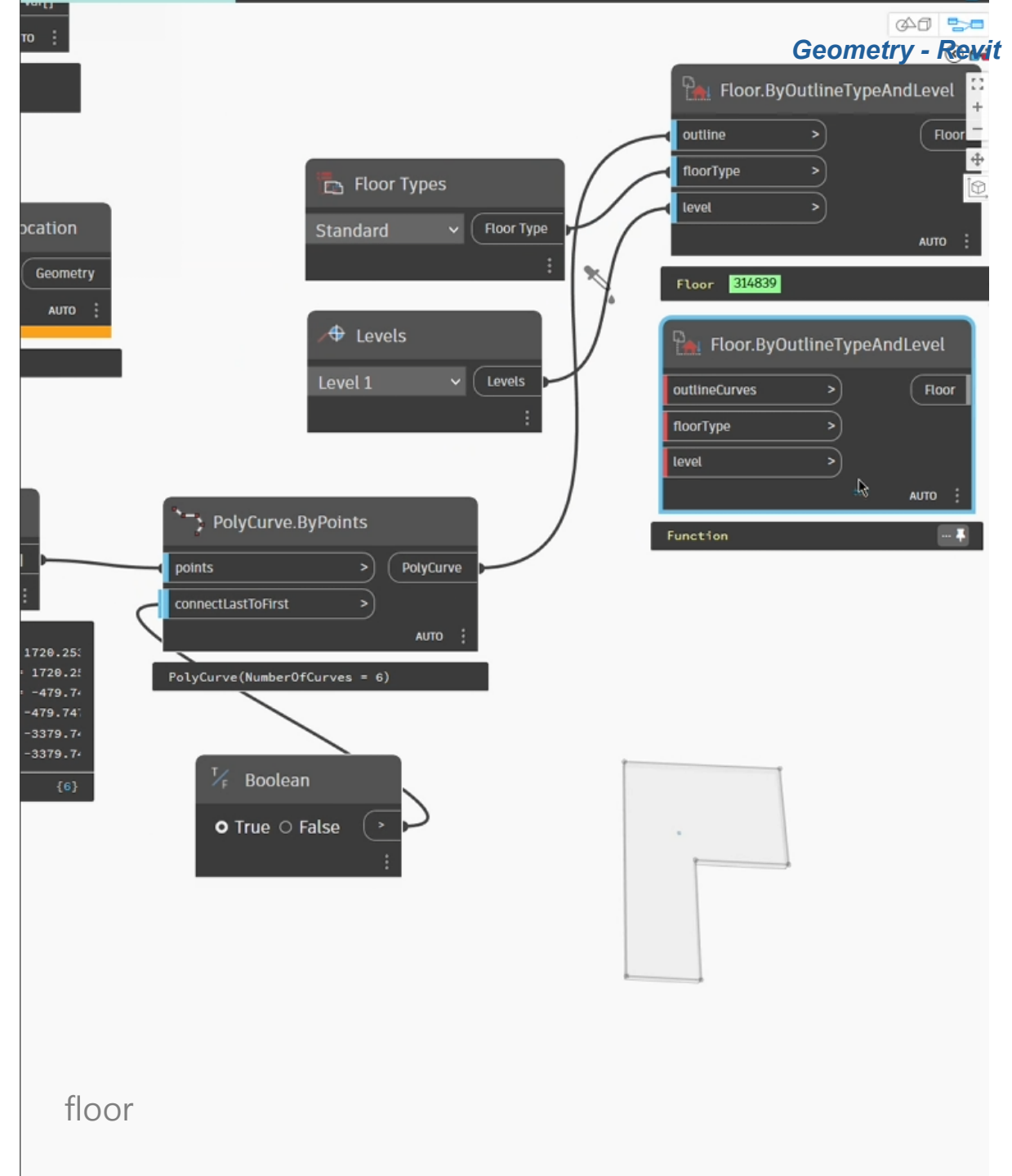
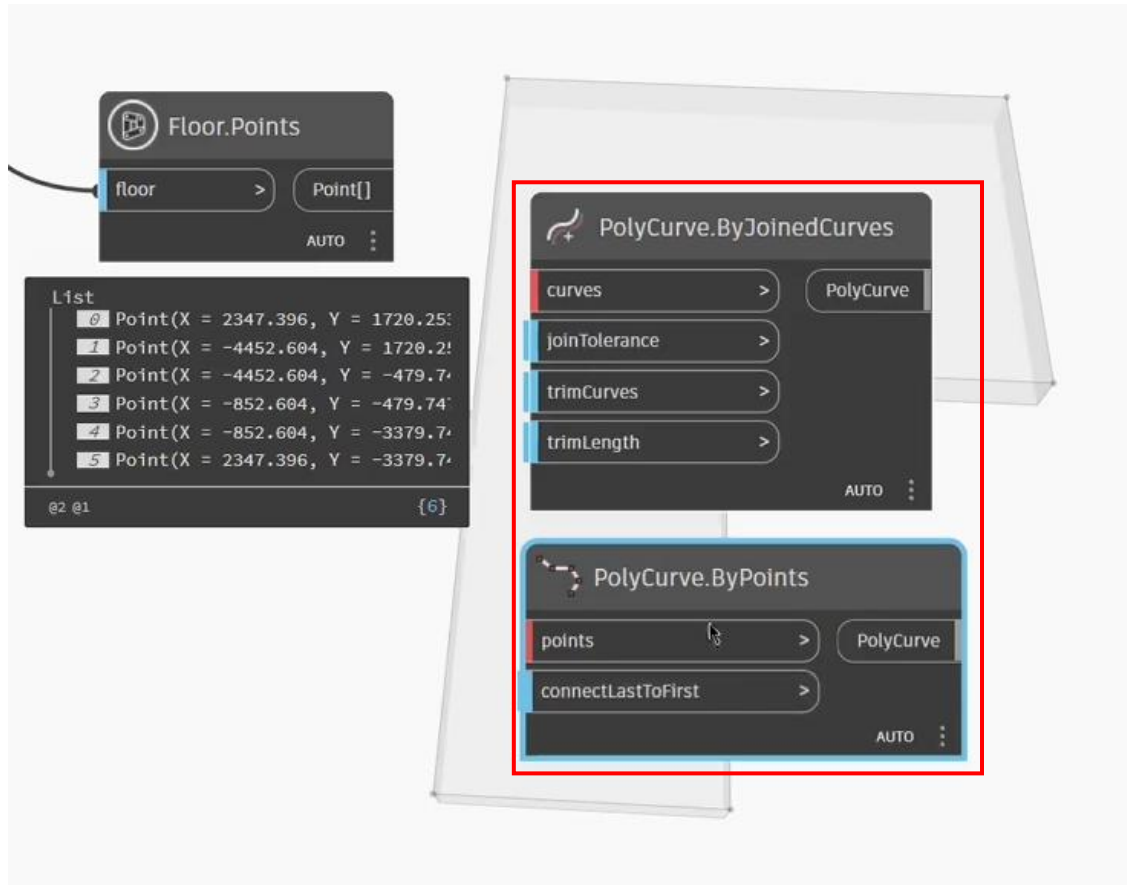
AUTO

```
List
> 0 List
  0 Line(StartPoint = Point(X = -
> 1 List
  0 Line(StartPoint = Point(X = -
> 2 List
  0 Line(StartPoint = Point(X = 3
> 3 List
  0 Line(StartPoint = Point(X = 3
> 4 List
  0 Line(StartPoint = Point(X = -
> 5 List
  0 Line(StartPoint = Point(X = -
> 6 List
  0 Line(StartPoint = Point(X = 2
@3 @2 @1 {8}
```

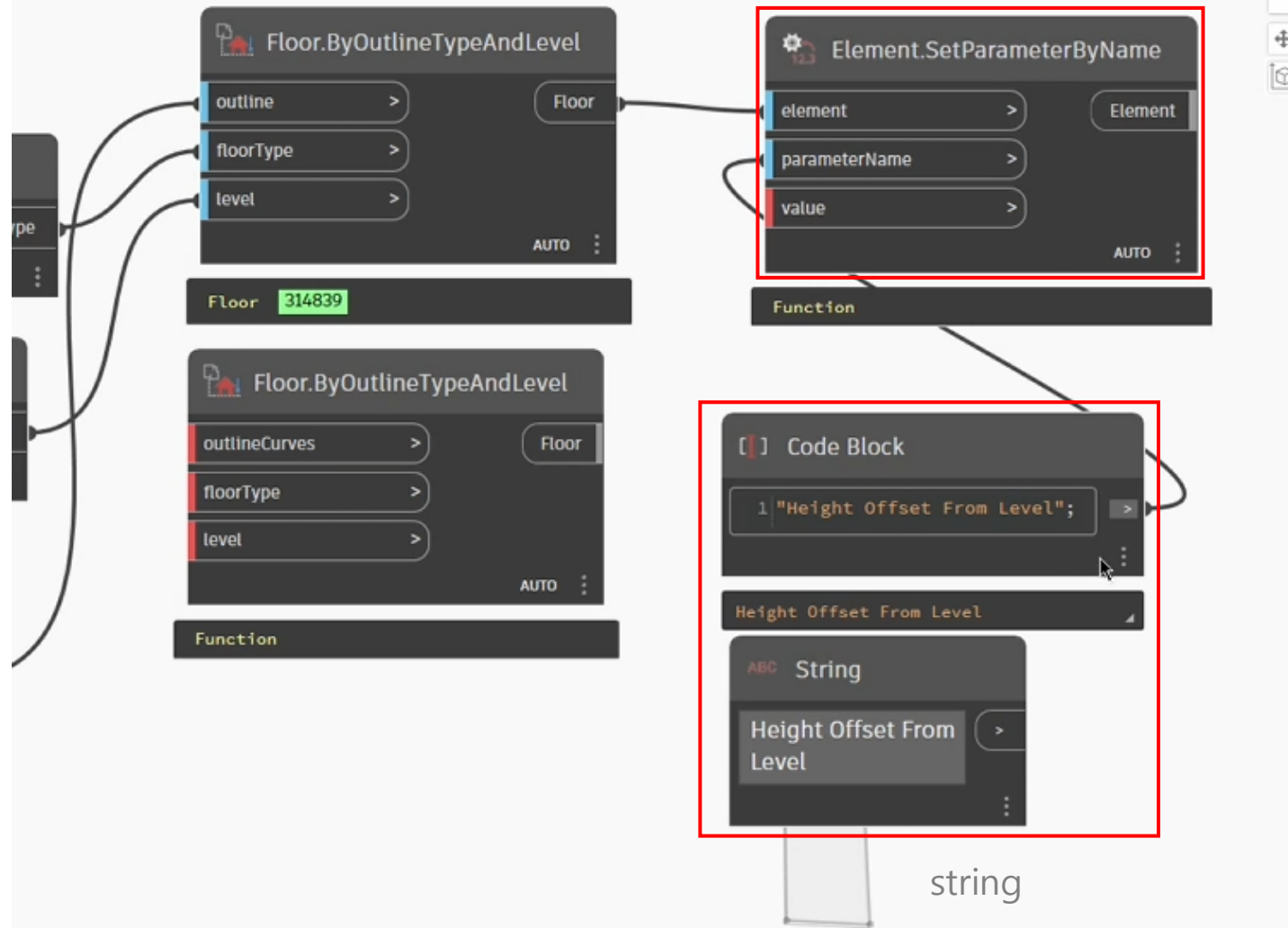


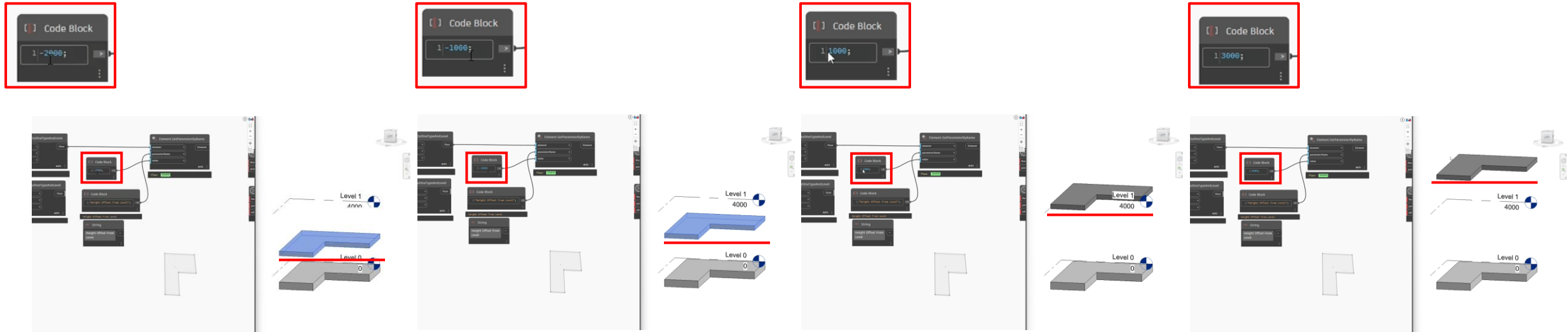


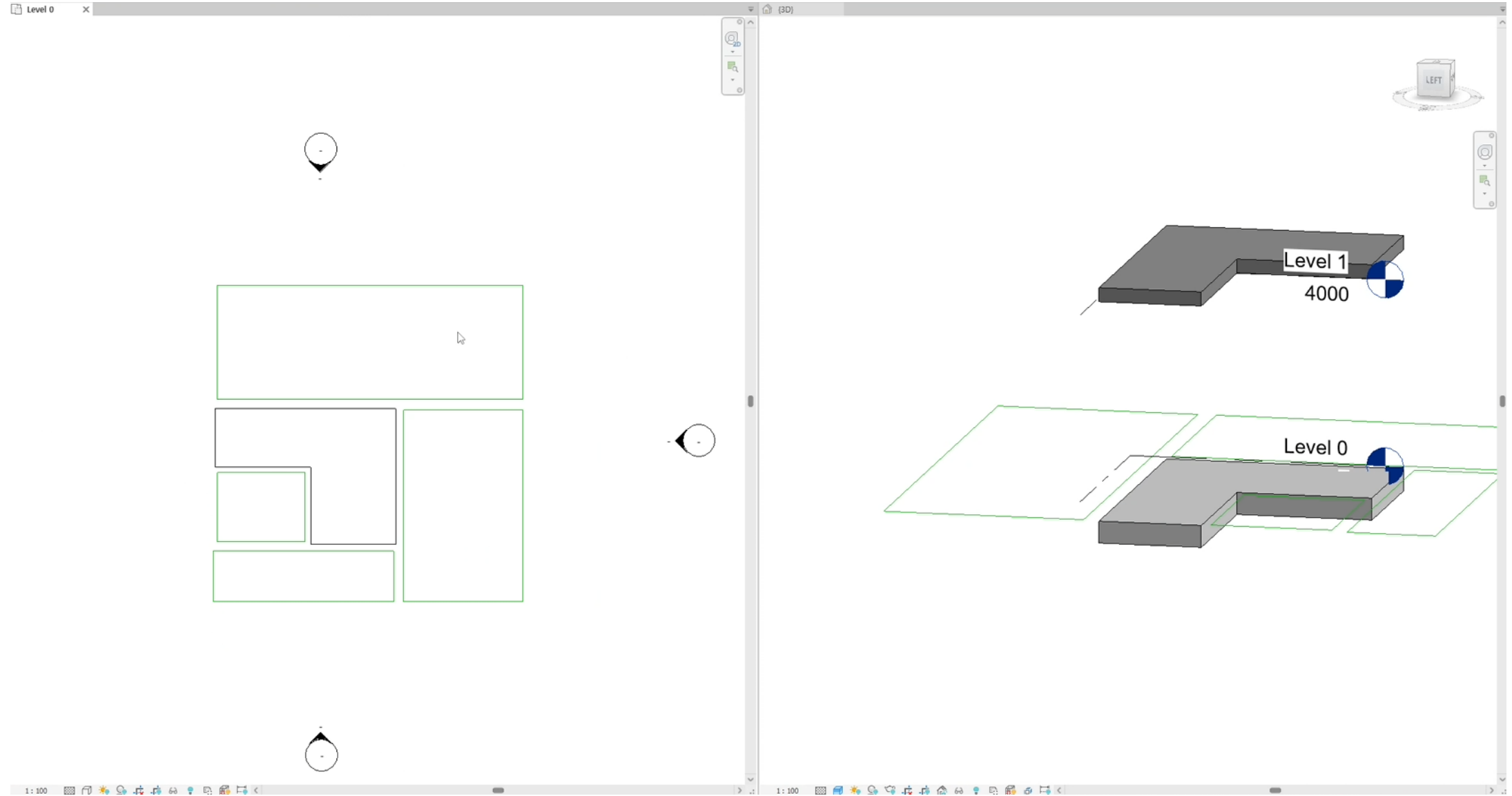
Select model elements
Floor.Points
polycurve

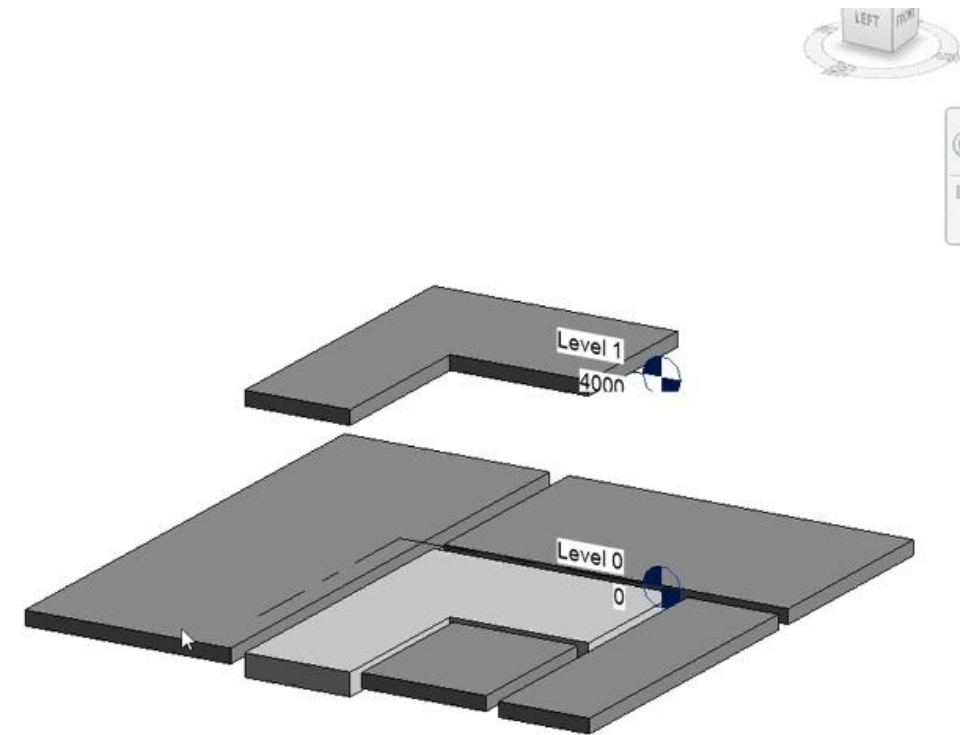
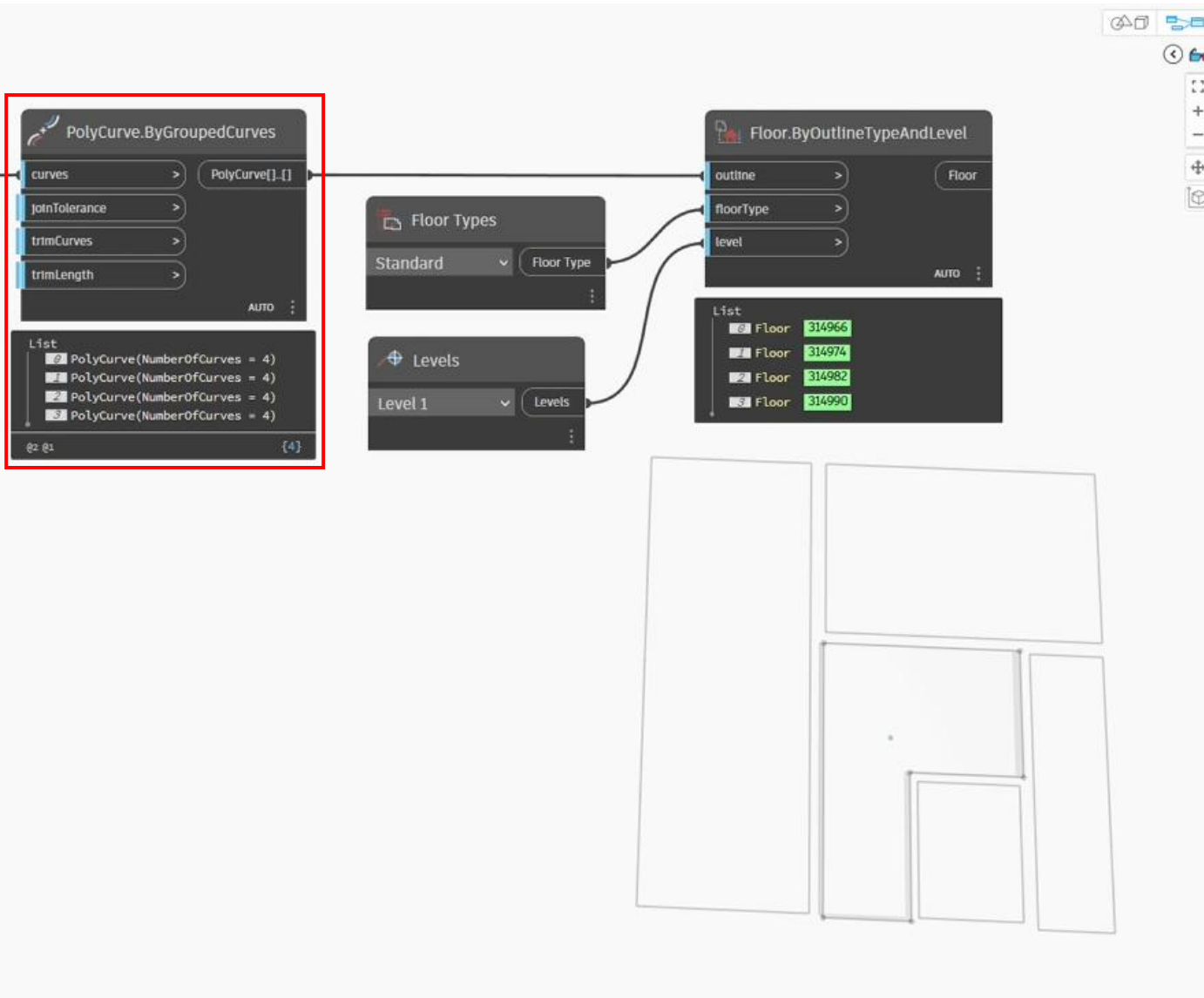


setParameter









Next